

Beyond Getting Work Done: A Landscape Review of Semantic Scaffolding for Learning with AI

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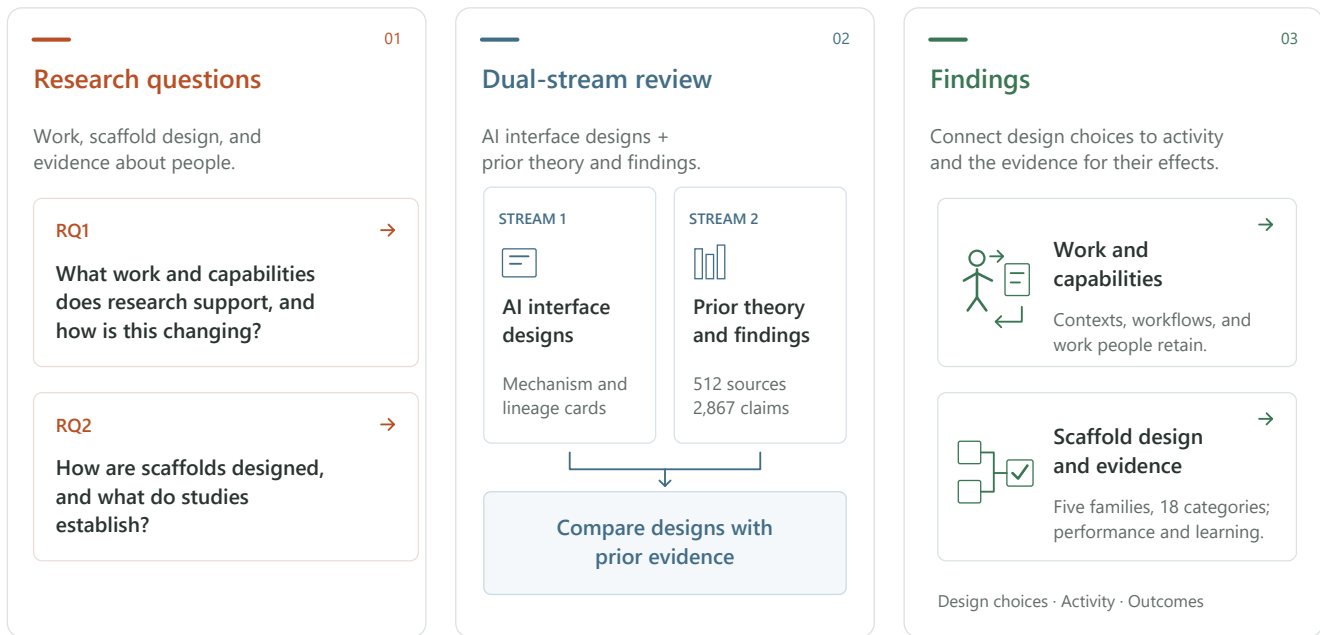


Figure 1: Work, capabilities, and scaffold design. Two research questions guide a dual-stream review of AI interface designs and prior theory and findings. The comparisons examine how features work together, what systems provide and people still do, and what earlier studies tested. The Stream 2 panel shows the synthesis selection: 512 sources and 2,867 claims after source and claim checks.

ABSTRACT

AI tools now help people write, program, analyze data, and design, and many also aim to help people learn through that work. What such a tool gives a person to think with varies: a generated explanation, criteria for judging alternatives, an editable plan, a record of earlier choices. We call these designed structures with task meaning *semantic scaffolding*. We reviewed 4,511 AI-involved papers from 40 venues (2019–2026), together with theories and empirical studies of how support works, and synthesized a design space of 5 mechanism families and 18 categories. Compared with papers on earlier AI, generative-AI papers more often concern producing artifacts and more often give people control over what the system does, after adjusting for venue and year. What a person can edit matters: the task specification, the AI’s interpretation of it, the criteria, or the result. Designs also differ in whether people construct a representation, reason with supplied material, or judge proposed results. Studies that measure learning report gains and costs together: CodeVoyager raised closed-book comprehension without a completion advantage, and required reflection cut completed practice from 3.4 to 2.0

problems in a fixed time. The design space gives researchers named alternatives to build and test.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**; *Graphical user interfaces*; *Collaborative interaction*; *Natural language interfaces*.

KEYWORDS

Semantic scaffolding, learning, human–AI interaction, external cognition, landscape review, interface mechanisms

1 INTRODUCTION

A writer asks Luminate for story ideas. Instead of one draft, the system proposes dimensions along which ideas can differ and fills each with options [119]. The dimensions help the writer survey the possibilities and compare ideas. They are also generated, and a generated dimension can miss what matters to this writer. The

writer now judges two things: the ideas, and the structure offered for judging them.

This is the design problem we study. AI tools increasingly supply the structures people think with, not only the content they produce: criteria, examples, relations between concepts, plans, and records of earlier choices. We use *semantic scaffolding* to describe designed structures with task meaning that a person inspects or acts through. The term joins two lines of work. Scaffolding research asks how support lets a learner do what they could not yet do alone [99, 132]; cognitive-artifact research asks how representations change the task a person performs [91]. The assistance dilemma and ICAP add that what support supplies, and what it leaves for the learner to do, shapes what people learn [27, 62].

Broad labels such as feedback, reflection, and user control hide the choices that matter. Editing a generated answer differs from correcting how the AI interpreted the request before it answered. A learner's account prepared for a coach differs from a simulated learner used to rehearse coaching. We map these choices across domains such as programming, writing, data analysis, and design, connect them to the work people do, and compare what studies found when the choices were tested.

We reviewed papers from 40 HCI, AI, and learning venues published from 2019 to 2026. Working with language models, we screened 35,480 titles and abstracts, selected 7,372 papers for full-text review, and analyze 4,511 in which AI is part of the system or study and a person inspects or acts through a designed structure. A second body of work, theories and empirical studies of how support works and when it helps, gives the comparisons we use to interpret designs; it includes studies without AI. We ask:

- **RQ1.** What work and human capabilities does semantic scaffolding research support, and how has this focus changed from 2019 to 2026?
- **RQ2.** How is semantic scaffolding designed, and what does existing evidence show about its support for performance and learning?

We contribute:

- (1) a design space of semantic scaffolding in AI-assisted work, with 5 mechanism families and 18 categories, each illustrated by named systems;
- (2) an account of what people can inspect and change in these designs (the specification, the AI's interpretation, the criteria, or the result) and of how support passes between learners, peers, teachers, and coaches;
- (3) a comparison of the consequences studies report for performance and learning, tied to the design alternatives they tested.

2 BACKGROUND

2.1 Scaffolding and Cognitive Artifacts

Scaffolding began as an account of tutoring: a tutor takes over the parts of a task a learner cannot yet manage, so the learner can do the rest [132]. Developmental scaffolding responds to the learner, reduces help as the learner improves, and hands over responsibility [123]. Reiser separates structuring a task from problematizing it, drawing attention to what the learner must reason about [104],

and inquiry-support guidelines turn these ideas into design [103]. Support can also be spread across people, artifacts, and tools [10, 99]. Pea argues that support meant to stay should be kept apart from scaffolding that fades. Sherin et al. extend the metaphor into a comparative account of scaffolded activity [113], and Puntambekar and Hübcher warn that calling a tool a scaffold does not make its help responsive to the learner [102].

Cognitive artifacts are support meant to stay. Norman uses the term for designed devices that store, display, or transform information for a task [91]. He distinguishes what the person and the artifact accomplish together from how the artifact changes the person's task: a checklist moves work into preparing, consulting, and interpreting the list. The same content supports different operations when it is arranged differently [109, 142]. People also change the world to help themselves reason [60], and several representations can aid understanding while adding the work of coordinating them [2].

We use *semantic scaffolding* for designed structures with task meaning—concepts, relations, examples, criteria—that a person inspects or acts through, whether they fade or stay. Zong et al. use the term for an LLM technique that explains domain groupings of data to blind and low-vision readers [147]; we use it more broadly. Keeping the two kinds of support apart matters for evaluation: performance with a tool and capability without it are different outcomes, and only the second shows growth toward independence [99, 102].

2.2 Representations in AI Interfaces

Recent design spaces describe these representational choices directly. Min et al. separate an abstraction's schema from its representation and examine granularity, presentation, and transformation [86]. Kim et al. organize how generative interfaces elicit preferences along six axes, including when a request appears and which part of an artifact it concerns [58]. Systems show the range. In Graphologue, selecting a concept extends a diagram with related concepts and links the person can follow and question [54]. In AI Chains, intermediate steps expose how a problem, a suggestion, and drafted feedback connect, and people can change each step [135]. In HaLLMark, a highlight marks authorship and links a passage to the prompt that produced it [47]. A cue can show that an action exists without showing what it does [92]: in Codestrates v2, a file-like tree organized code, and its order also governed execution, which one participant discovered only in use [18].

People and technology change together. Mackay describes users who repurposed email filters, built to sort messages before reading, into tools for filing them afterwards; the developers then supported both uses [80]. Interfaces have long adapted controls to a person's abilities [40, 130]. LLMs add support generated during use: diagrams that grow during exploration [54], views people recompose [85], and representations generated as tasks change [23]. A generated view can perform an operation, expose relations, and become the control for the next step. The design questions are who builds it, who can revise it, and what work its use adds or removes.

2.3 Offloading and Control in AI-Assisted Work

Distributed cognition treats people, tools, and artifacts as one cognitive system [45]. The extended-mind argument counts reliably consulted resources as part of memory [30], and material structures hold relations otherwise kept in mind [53]. Offloading changes what people retain: expected access to information shifts memory toward where to find it [116], and automation can leave operators with the hardest cases and little practice [8]. Mixed-initiative interaction combines automation with direct manipulation [48]. Interactive machine learning makes the user–system loop the object of design [3], explainable-AI frameworks separate understanding, trust, debugging, and decision support [89], and teaming and learning-support research adds capability and transfer [22, 42]. Because people trade accuracy against effort [98], making some operations easy also changes strategies. Generative systems that expose prompts, plans, alternatives, and evidence [24, 61, 119, 134] also produce task criteria and explanations on demand, so people may need to check the guidance as well as the answer.

2.4 Landscape Reviews and Design Spaces

Design spaces organize recurring choices and the criteria for weighing them [81], and how they are drawn shapes what designers explore [14, 68]. HCI landscapes relate creativity support tools to practice and evaluation [36], organize traces of creative activity [44], and examine generative tools in artists’ work [61]. Citeology, Royal Constellations, and HCI Research Trails make citation relationships navigable [19, 84, 88]; we also record what a cited source does in a paper: frame it, explain a mechanism, shape a design, or provide a measure.

Earlier syntheses relate contingent support and learner activity to learning [11, 123]. Self-explanation research identifies practice exposure and redundant prompting as conditions that matter [106]. The assistance dilemma and ICAP frame when to give help and what engagement it invites [27, 62], and expertise reversal and desirable difficulties qualify what a practice result means [15, 56]. Krakauer distinguishes artifacts that leave capability behind when removed from those whose advantage lasts only while they are in use [63]. Human–AI guidelines already address intelligibility and correction [4]. Our addition is a comparison of concrete design choices across domains, linked to the activity each asks of people and to what studies measured when they tested it.

3 METHODS

We reviewed two bodies of work (Figure 2). Stream 1 is a systematic corpus of AI-involved systems and studies from fixed venues and years; it supplies the designs we map. Stream 2 collects theories and empirical studies of how support works, including studies without AI; we interpret the designs against it. The two are counted separately. Scoping-review guidance shaped selection, extraction, and reporting [6, 69, 121], and earlier HCI landscape reviews shaped the venue frame and synthesis [36, 44].

3.1 Literature Review

3.1.1 Venues. Stream 1 covers 40 venues in HCI, design and creativity, visualization, graphics, learning and education, software engineering, and AI and language, published from 2019 to 2026

(Figure 3). Papers are dated by their version of record; the selection figure, the design space, and the trend figures cover papers published by 16 July 2026. Ineligible publication types were excluded by rule, and citation counts played no part.

3.1.2 Screening. Two language models separately screened the 35,480 records with an abstract; the 2,498 without one were not screened. Their verdicts, a review of their disagreements, and (for the first reading list) PDF availability determined the 7,372 records selected for full-text reading. One model read each paper and recorded, with passages and pages, whether it met the criteria.

The AI requirement was added partway through full-text reading, so three separate model passes rechecked the 1,154 earlier cards that left the question open. A paper left the corpus only when all three found no AI, with low uncertainty and a quoted passage (492 cards). The corpus holds 4,511 papers, and 1,076 of them form a learning subset: their support has a documented learning purpose, or the study examines its learning consequences. Another 474 papers without AI form a labeled comparison set outside the systematic corpus counts; the text marks any it uses as an example.

3.1.3 Inclusion Criteria. Stream 1 includes a paper when three conditions hold, at screening and at full text (Figure 4): AI is part of the system or the study; the paper implements or studies a designed structure with task meaning; and a person inspects or acts through that structure. AI is part of a paper when an AI system is built, used, evaluated, or simulated (including Wizard-of-Oz systems presented as AI), or when AI is what the study designs, explains, or labels. Rewording a chatbot’s replies (hedges, tone, length) is not a designed structure. Workshop templates, cards, storyboards, and probes used to elicit views count only when the paper builds and evaluates them as its contribution. Reviews, surveys, and meta-analyses go to Stream 2. A design can qualify without a user study; its benefits then count as intended uses, not measured effects. Learning purpose, measured outcomes, adaptation, fading, and the transfer of responsibility are recorded but do not decide inclusion.

3.1.4 Theories and Empirical Studies. Stream 2 gathers theories, design principles, and empirical findings about how support works. Its sources came from database searches, backward and forward citation snowballing [131], and reviews found in Stream 1, and every source received the same full-text assessment of relevance and support. Of 795 candidate works, we read 543 in full and excluded 31 for recorded reasons, such as falling outside the review questions or appearing in a questionable venue. The analysis uses the remaining 512 sources and their 2,867 claims. Books, foundational theory, null findings, and studies without AI are eligible.

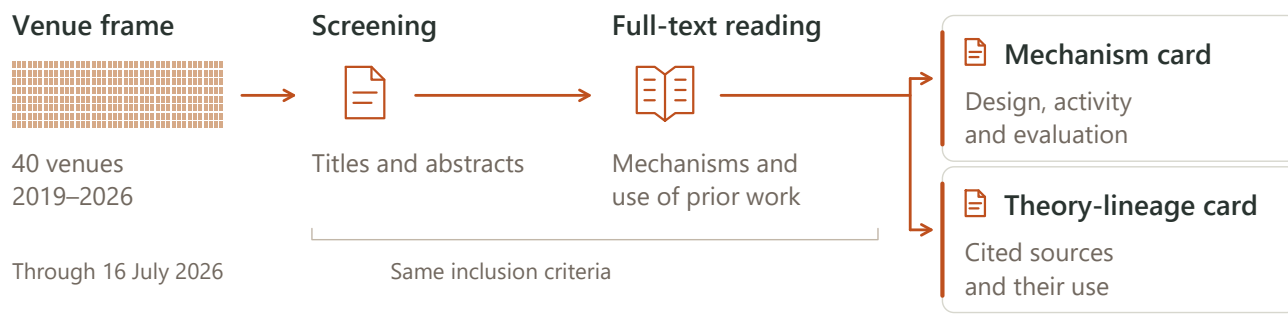
3.2 Analysis

3.2.1 What We Recorded. A model read each full text and filled a structured evidence card, quoting the passages and pages behind its entries.¹ Scientific-document benchmarks and expert-assisted literature mining offer precedents for such passage-linked records [124, 128]. The Stream 1 unit is a *mechanism*, an operation involving a designed structure with task meaning; AI Chains’ workflow overview and its step editor are two mechanisms [135]. For each

¹Evidence cards: mechanism, theory-lineage, and knowledge-claim cards.

Literature selection and evidence extraction

Stream 1 · AI-involved designs



Stream 2 · Prior theory and findings

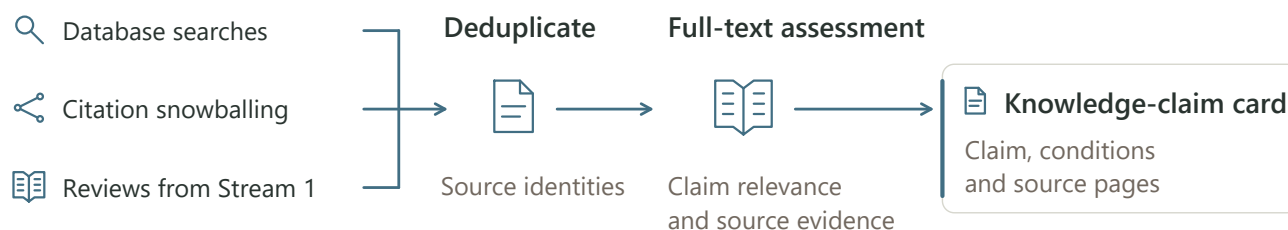


Figure 2: Literature selection and evidence extraction. Stream 1 runs from a venue frame through screening and full-text reading to mechanism and theory-lineage cards. Stream 2 combines three discovery routes, removes duplicates, and records each source’s claims on a knowledge-claim card.

mechanism, the card records what the interface represents, how and when it presents that meaning, and what people can do. It also records how the work changed, what the tool took over and what the person kept, and how the design was evaluated: comparison, measures, duration, participants, and results. A second card records how the paper uses each cited source: to frame a problem, explain a mechanism, motivate the design, inform a measure, critique an approach, or in passing. The Stream 2 unit is a *knowledge claim*, a finding or theoretical argument about how support works, for whom, and when; each card holds one to eight claims with their evidence, conditions, and limits. Missing information is recorded as not reported or unclear, and author claims, measured results, and reviewer inferences stay apart (fields in Appendix A).

3.2.2 Building the Design Space. Closed fields take predefined values, such as whether learning was measured and for how long; open fields describe mechanisms and human work in each paper’s terms. We merged the 142,877 open-field observations into 1,290 codes and grouped these into 223 categories and 92 families across 18 fields (Appendix A.3). Each grouping links to its observations, so the grouping and the assignments can be checked separately [50]. The mechanism field’s hierarchy is the design space: 5 families and

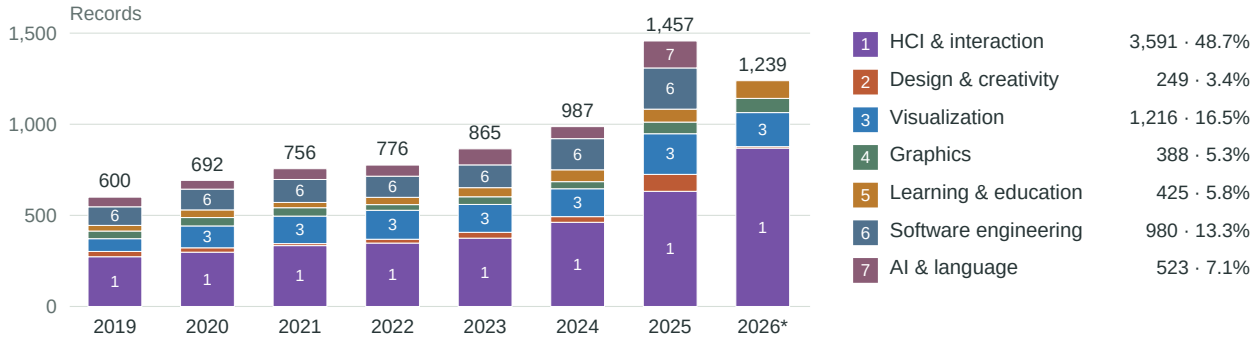
18 categories (Section 5; codebook in Appendix B). For the four-function view (Figure 11), we assigned each of the 110 mechanism codes to the functions it serves; exploring alternatives serves both representing and evaluating.

We built the vocabulary from the cards completed by 28 August 2026 and coded later cards with it. The design-space counts describe the 2,847 papers of that first set still in the corpus (2,891 for the closed fields); the annual and technology figures draw on the 4,086 papers with usable coding (Appendix A.6). A paper counts once per category, however many mechanisms it has. We classified each paper’s AI as generative or traditional AI/ML by the technology it used, not its year, and adjusted technology comparisons for year and venue, since venue mix alone can create differences (Appendix D).

3.2.3 Comparing Designs and Studies. We used the codes to find cases and the source passages to compare them. For each question, we compared the task, the representation supplied, the actions available, the reported use, and the assessed consequence, recording contrary and unclear cases beside supporting ones. We chose examples that fit a mechanism, varying domain and year; citation counts mattered only after fit. We retrieved prior findings and counterexamples through the seven fields coded on Stream 2 claims,

Selected records by year and venue group

7,372 records selected for full-text reading; venue groups describe disciplines, not application domains.



Each venue: selected records and share of the 7,372 selections

1 CHI	1,548 · 21.0%	1 ECSCW	10 · 0.1%	2 Design Studies	13 · 0.2%	5 AIED	58 · 0.8%
1 UIST	306 · 4.2%	1 GROUP	6 · 0.1%	2 IDETC	151 · 2.0%	5 IJAIED	37 · 0.5%
1 IUI	359 · 4.9%	1 ASSETS	84 · 1.1%	3 VIS	394 · 5.3%	5 EDM	103 · 1.4%
1 DIS	232 · 3.1%	1 HRI	187 · 2.5%	3 EuroVis	111 · 1.5%	6 CHASE	55 · 0.7%
1 CSCW	258 · 3.5%	1 IVA	82 · 1.1%	3 TVCG	635 · 8.6%	6 ICSE	321 · 4.4%
1 C&C	67 · 0.9%	1 CHIIR	56 · 0.8%	3 VL/HCC	76 · 1.0%	6 FSE	238 · 3.2%
1 TOCHI	45 · 0.6%	1 HCOMP	36 · 0.5%	4 SIGGRAPH	295 · 4.0%	6 ASE	366 · 5.0%
1 TiiS	102 · 1.4%	1 IJHCS	127 · 1.7%	4 SIGGRAPH Asia	93 · 1.3%	7 ACL	191 · 2.6%
1 EICS	33 · 0.4%	2 ICC	53 · 0.7%	5 L@S	41 · 0.6%	7 EMNLP	255 · 3.5%
1 HAI	53 · 0.7%	2 Design Science	32 · 0.4%	5 LAK	186 · 2.5%	7 RecSys	77 · 1.0%

*2026 is partial: publication cutoff 16 July. Zero selections can reflect unavailable proceedings.

Figure 3: Records selected for full-text reading, by publication year and venue group. The lower panel gives each venue's count and share of the 7,372 selections. The 2026 bar covers papers published by 16 July.

such as the form of support and a claim's conditions. The lineage cards trace how a cited theory or finding entered a design or a measure (Section 5.5); reference lists and citation counts come from Semantic Scholar (Appendix A.9.2).

For study results, each contrast keeps its population, comparator, outcome, timing, and the assistance available during assessment, and positive, mixed, null, harmful, and unmeasured results stay distinct. A whole-system result is credited to the whole system, and a learning claim counts as tested only when a study measured learning or transfer. We sorted outcomes into process, artifact, and learning, each assessed by the study or reported by participants (Section 5.6). We did not pool effects across studies (Section 8).

3.2.4 Checking the Coding. Scripts checked the cards' required fields and matched their quotations against the source text, and quotations the text layer could not match were checked against page images; counts and plots are computed from the saved data. To test the open-code grouping, a separate model pass coded 1,571 extracted observations in six fields with the existing codes hidden. Agreement was low (adjusted Rand index .079 to .445), and the blind pass grouped more coarsely in five of the six fields (Appendix A.4). The mechanism vocabulary gained 12 codes across the 1,754 papers read after it was built (Appendix E).

3.3 AI Use Disclosure

We used language models to screen, read, and code papers, group codes, develop and critique the cross-case comparisons, write analysis code, and draft and revise text, tables, and figures. Claude Opus 5 and GPT-5.6 Sol screened titles and abstracts. GPT-5.6 Sol, Claude Opus 5, and Claude Opus 5.5 read the full texts and wrote the Stream 1 cards; Claude Opus 5 and GPT-5.6 Sol wrote the Stream 2 cards; Claude Opus 5.5 ran the three-pass AI check. Claude Opus 5 merged labels into codes, and model-assisted review, including Claude Table 5.1, grouped the codes into categories and families. GPT-6 Sol linked card evidence for the cross-case comparisons, and GPT-6 Astra developed and critiqued them. GPT-6 Astra, working through Codex, drafted the merged manuscript, and Claude Opus 5.5 revised it. Prompts and run records are kept. The authors set the research questions, scope, criteria, and card formats, directed the synthesis, and are responsible for its conclusions; people have not yet checked the coding or the cross-case interpretations (Section 8).

Inclusion criteria

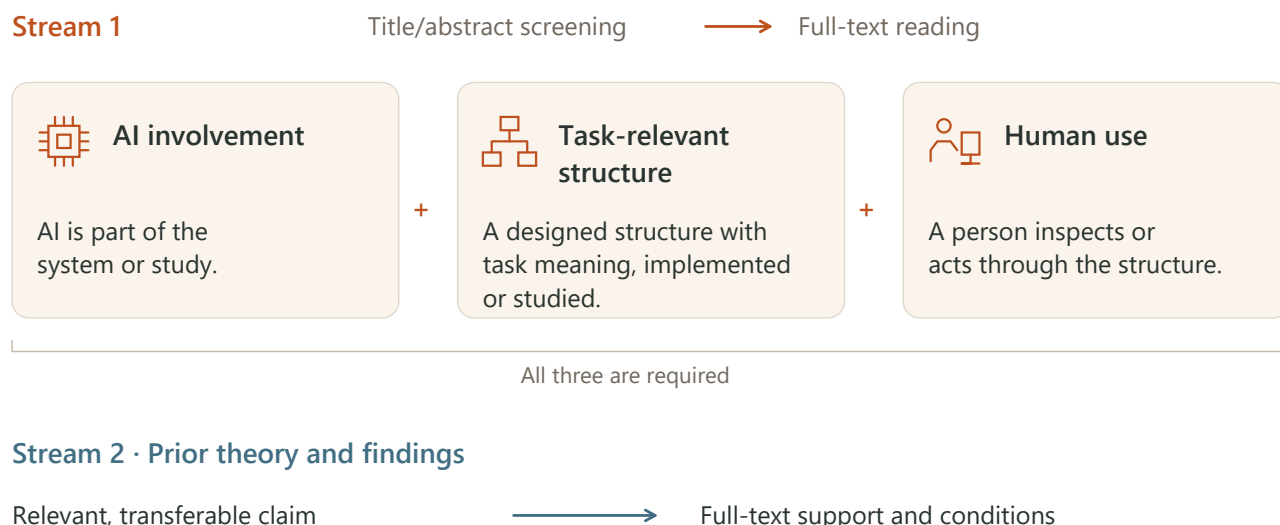


Figure 4: Inclusion criteria. Stream 1 requires all three conditions at screening and full text. Stream 2 assesses whether a claim is relevant and supported by its full text. Learning purpose and measured outcomes are recorded during extraction.

4 FINDINGS: WORKFLOWS AND CAPABILITIES

GenAI-classified papers more often describe creating, editing, and repairing artifacts. This shifts attention to the material people encounter and the decisions they still make. We first place the shift among the work domains papers study, then ask what the person still composes, checks, and decides, and last how support passes through other people.

4.1 Work represented and changes in research attention

Within venues, we detected no annual change in a domain’s share from 2019 to 2025 after correction for multiple tests (Table 11). The quantitative pool contains 4,086 AI-involved paper groups; each comparison uses the subset with the required coding and technology classification. GenAI includes earlier generative models and generative hybrids, and traditional AI/ML appears in every year. GenAI becomes the majority from 2024 (Figure 5). Technology comparisons leave out the 639 papers whose technology is unresolved, 2026 is partial because papers were collected in July, and a paper can support several activities.

Composition explains some apparent technology differences. Programming accounts for 5.5% of GenAI papers and 7.8% of traditional AI/ML papers when years are combined. Adjusting for year and venue in the same papers gives +0.9 percentage points (95% CI -0.8 to 2.6); the 2019–2025 estimate is similar. The activity comparison is clearer: across 2019–2025, creating, editing, and repairing artifacts is more common in GenAI papers (+11.0 points;

95% CI 5.6 to 16.4). Its direction and Holm-adjusted support persist in both sensitivity windows (Appendix D.2).

Appendix D gives the annual domain and activity distributions with the paper counts behind each share (Figure 54 and Figure 55). The paired workflow figure uses 2,390 papers with an activity code and assigned technology. Allocation, supplied support, reported outcomes, and study-description codes remain separate: faster work, for example, does not establish automation.

4.2 Delegating work while retaining judgment

Two choices shape what the person still does: how much of the material the system has already composed, and what the person can check it against. CorpusStudio and ABScribe both support writing but differ in the first. CorpusStudio supplies examples and patterns from prior writing, which authors interpret before composing their own prose [31]. ABScribe generates alternative passages that authors keep, edit, and compare [105]. One gives authors material to write from; the other gives them passages to select and revise.

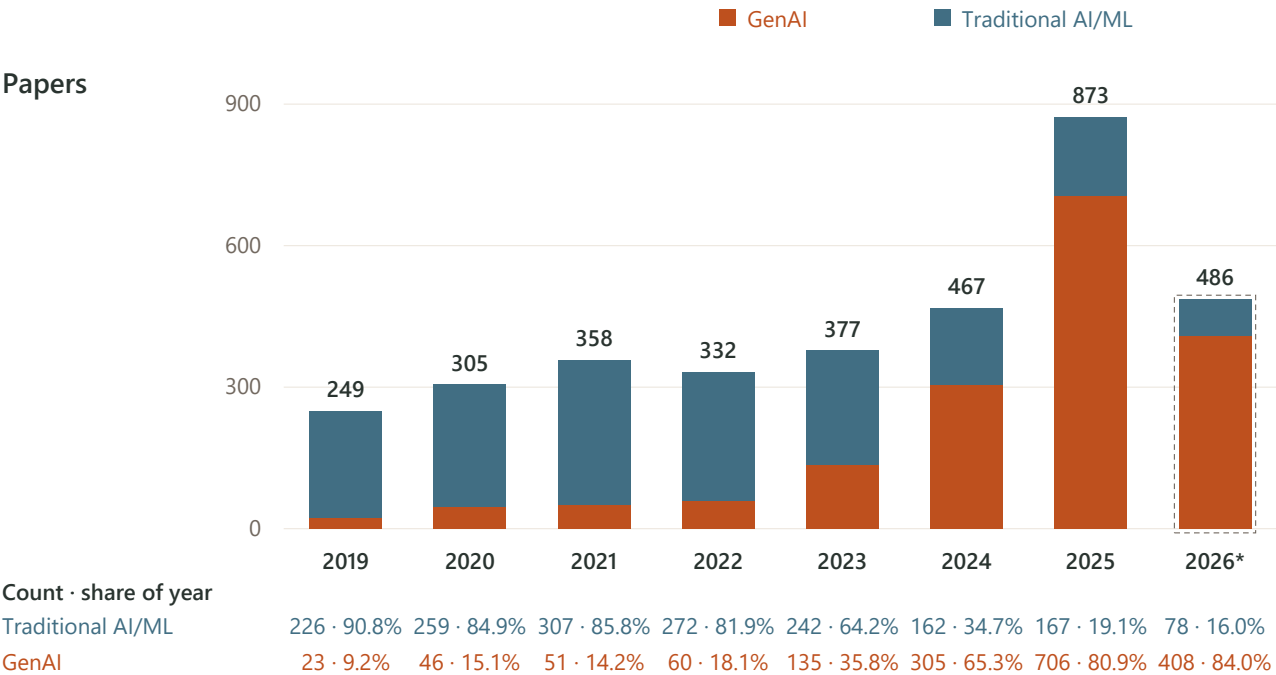
Figure 6 organizes the human activities: judgment, doing and revising, checking, and responsibility or self-directed work. These categories locate the action to examine in each design.

Norman separates what a system accomplishes from the activity of the person using it [91]. In the reviewed interfaces, composing a passage, choosing a criterion, inspecting execution, and deciding whether to keep an alternative are separate operations, and designs divide them differently.

Different references support different checks. A proposed result can be checked against executed values, retrieved examples, the system’s interpretation of an action, or a generated forecast.

AI technologies by year

3,447 classified papers · bar height shows volume; percentages show composition



GenAI includes hybrids and earlier generative models. * 2026 through 16 July; September update pending.

Figure 5: AI technologies by year. Stacked bar heights encode paper counts, with annual totals above. Rows below give each segment’s count and within-year percentage. Classification follows the technology used; 2026 is partial (collected in July). The figure omits 639 papers without a resolved technology classification.

Leap shows the values that generated code produces as it runs [35]; CorpusStudio retrieves examples from a writing corpus [31]; ONYX shows how it interpreted an action the user taught it [108]; and Rehearsal simulates how a conversation partner might respond [111]. Each grounds a different judgment. A simulated response is material for rehearsal, not a validated prediction of what a particular person will say. A corpus supplies precedents, but its incomplete coverage cannot show that an idea is new.

A reference helps only when people notice and use it. ONYX’s full interface improved script accuracy, yet users missed or wrongly dismissed 44.8% of its follow-up questions [108]. Leap’s live values did not significantly change validation success compared with a condition in which people could still run the code themselves [35]. A design therefore has to name the error a person should catch, the information that reveals it, and the response that would show the check worked.

ExPeerience shows that generation need not turn learners into passive checkers: it uses AI to create problem contexts, tables, and explanations but leaves writing the SQL solution to the learner [145].

4.3 Coordination and opportunities to learn

Support can pass through another person whose judgment remains necessary. In entrepreneurship coaching, the novice describes a project and selects risks; the interface gives the mentor selected and omitted risks with source traces to prepare a meeting [51]. HTCT instead turns tutor discourse into analytics that a coach selects and discusses with the tutor [17]. One representation carries a person’s expressed priorities; the other supplies an analysis of their activity. The immediate interface user, contributor, and intended beneficiary can be different people.

These arrangements require following what crosses roles. The entrepreneurship study observed eleven novices and one mentor in real meetings, with changes in plans and diagnoses. Coaches chose whether to use HTCT. The study followed their tutors’ practice for seven weeks across early, delayed, and no-access groups. Both studies give evidence about coaching practice, not about what students learned from a coach’s interface. Rehearsal serves a different purpose: its generated counterpart provides a setting to practice conflict responses [111]. A simulated response is practice material, not evidence of how a real recipient understands.

What work remains with the person?

Read the human activity alongside what the system supplies.

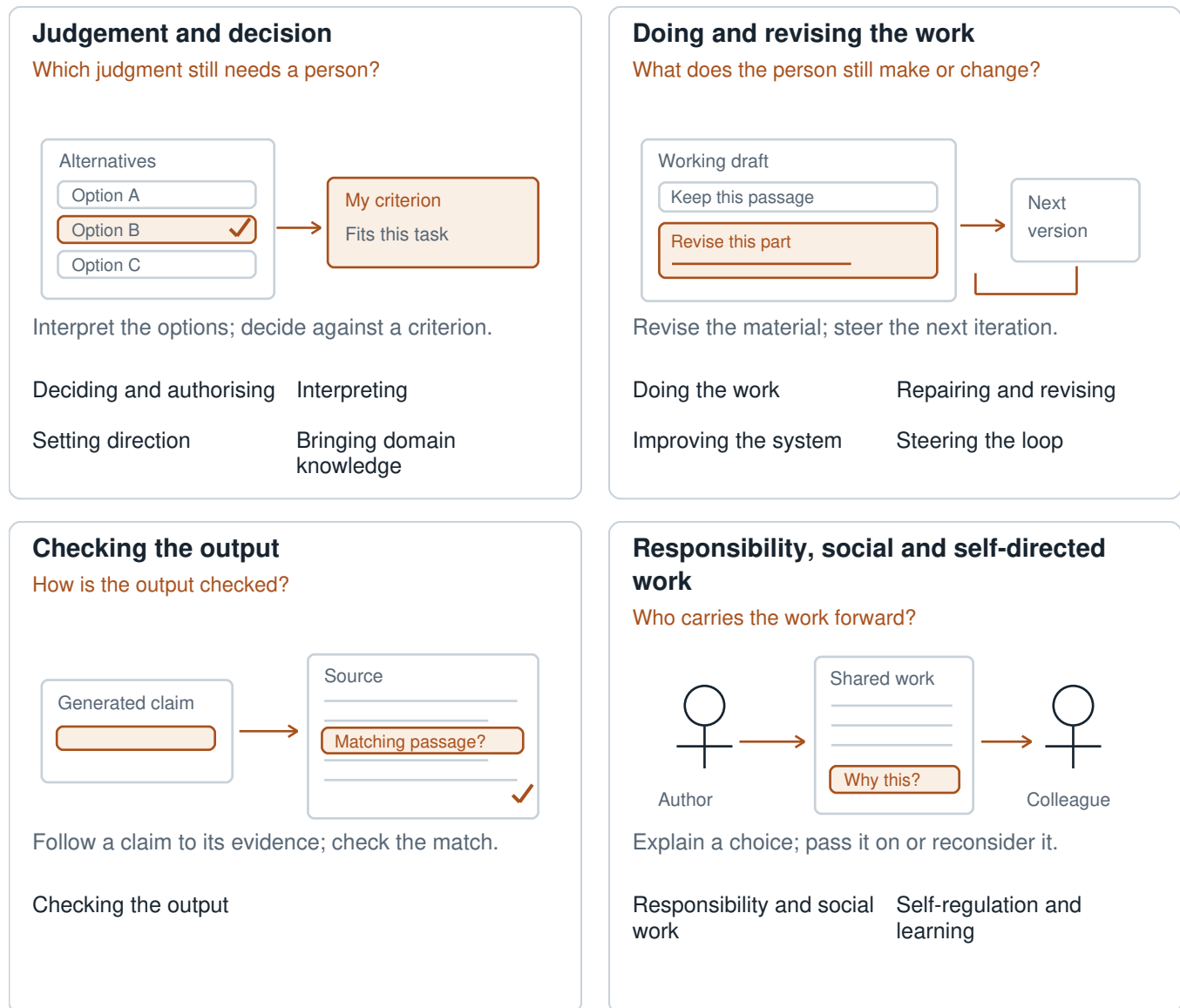


Figure 6: What work remains with the person? Four substantive families organize eleven categories of retained human work. Orange marks an intervention in these schematic examples. The coding also records when retained work is not stated; that reporting state is not an activity. The categories describe work, not measured uptake or learning. Browse the activity examples.

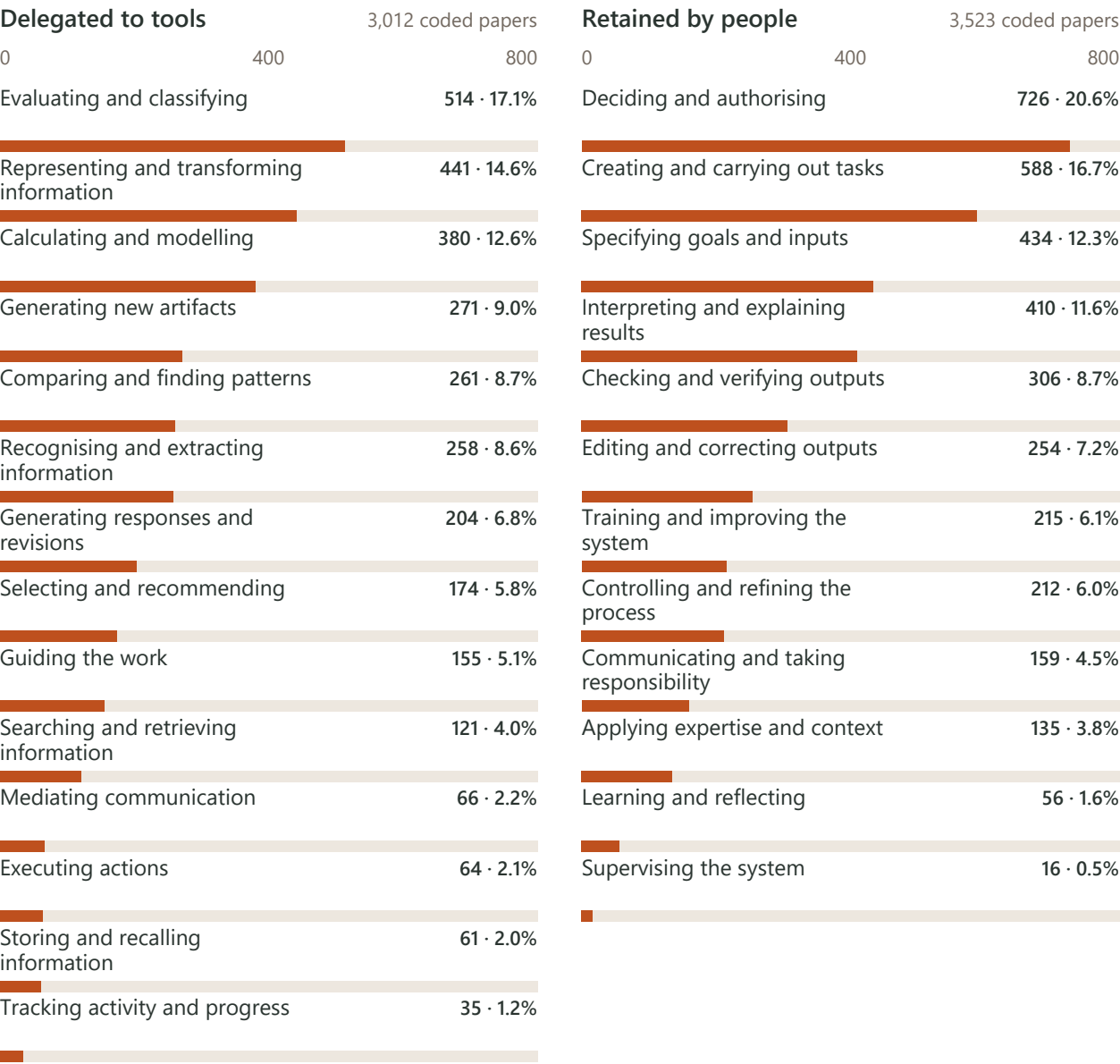
A shared representation does not guarantee a response. In eaSEL, a child's reflective artifact can be shown to a parent with a prompt about the parent's own experience, but the evaluation did not observe the whole parent-child exchange the paper proposes [112]. QMaps, a system without AI, makes the participation issue concrete. Students create questions, link prerequisites, and justify those

links. Incorrect links sometimes revealed copied content that students did not understand, while students preferred adding their own questions to revising others' work [137]. Exposing reasoning could reveal a difficulty without ensuring that anyone addressed it.

The social arrangement can also change when software supplies feedback. BrailleBlocks makes a child's construction interpretable to a sighted parent who can offer hints [38]. In WriteAid (Stream 2),

Work delegated to tools and retained by people

Bar length: number of papers · percentages use each field's coded papers



Counts use all papers coded for each field. Allocation conditions and unspecified work are listed with the data.

Figure 7: Delegated and retained work, ranked by prevalence. Bar lengths encode paper counts on a common scale; labels also give percentages within each panel's coding denominator. A paper can record several roles. Each panel uses papers with usable codes for that role; technology classification is not required.

some recurring questions became less visible to a teacher when pupils sought software feedback, while the teacher gained time to circulate and new peer conversations occurred [90]. These cases

turn on who has the information needed to help. A shared representation should be judged by the response it enables; visibility and collaboration are not learning outcomes in themselves.

Educator-facing support can also target different work. VizProg users diagnosed learners' code and selected feedback during a classroom replay, but that feedback was not delivered to students [141]. M-Powering Teachers supplied mentors with feedback about their own talk; randomized access increased their uptake of students' contributions and reduced their share of talk [32]. QLens lets designers inspect prospective feedback: one recommended path derived from successful traces was judged unsuitable as direct feedback for students [136].

These cases have different evaluation targets: diagnosis and feedback selection, ongoing teaching practice, and the suitability of proposed feedback. M-Powering's result covers the ongoing feedback package as a whole; the study did not record time on the feedback page and did not directly assess student learning. Educators can themselves be the intended learners. Evaluate the response the support is meant to change, and assess students separately when the claim is about student learning. QLens shows that a path students completed successfully can still be unsuitable as instruction.

The next section turns to the interface mechanisms that organize this work and to what evaluations of those designs show.

5 FINDINGS: MECHANISM TAXONOMY

User control is more common in GenAI papers, but the label conceals different design choices. People may revise a task specification, the system's interpretation, evaluation criteria, or the resulting artifact. Plume, for example, lets a writer preserve selected text while regenerating the surrounding explanation [71]. The object and persistence of an edit determine which decisions remain open to the person.

The 5 mechanism families and their 18 categories organize these choices. We describe the families and how their prevalence differs by technology, then compare concrete designs and what their evaluations establish.

5.1 Mechanism families

A mechanism pairs an operation with its form: a constraint can express a goal, an authored rule, or an adjustable threshold. Figure 8 organizes these forms, and Figure 9 compares their prevalence across technologies; a paper can use several. Table 16 counts the papers in each category and how many of them concern or measure learning, and the category maps in Appendix C.1 lead to particular mechanisms and their source papers.

The families describe different ways to support a decision. The *organization and comparison* family supplies alternatives, relationships, and histories. Graphologue turns text into entities and links [54]; Crystalline gathers options, criteria, and evidence from the pages a developer reads into a comparison table whose cells lead back to their sources [72]. The *guidance and feedback* family supplies recommendations, questions, and task structure. OATutor selects practice and reveals hints [97], while WithYou shortens the pauses it inserts into speech as a learner's shadowing improves [143]; revealing more help and reducing help are different interventions.

The *explanation and visibility* family exposes sources, rationale, or state. HaLLMark traces copied LLM text to its prompts but relies on writers to mark the passages they consider AI-influenced [47];

automatic provenance captures direct reuse more readily than inspiration. The *user control and expression* family lets people act on the material: Plume can lock text while it regenerates the explanation around it [71]. The *collaboration and perspective* family makes interpretations available to other people: Wikibench supports discussion that revises label definitions and inclusion rules [65].

To use the taxonomy, start from a decision the person must make. Find the mechanism that supplies information or an action for it, then ask what response it requires and which outcome would show success. A provenance link, for example, gives a basis for checking a claim, but the person still judges whether the source supports the wording. The link shows traceability; accurate checking, or learning to judge sources, needs its own observation.

5.2 Mechanisms across workflows and years

Control is more common; its components depend on which papers are compared. Among papers with usable coding, GenAI-classified papers more often describe user control and expression (+13.7 percentage points after accounting for year and venue; 95% CI 8.6 to 18.7). On the same 2,022 papers in shared year-venue cells, the gap is +12.5 with overlap weights and +7.8 when venues receive equal weight. Candidate choice is more sensitive to weighting: on its 2,016 shared-cell papers, the corresponding gaps are +7.2 and -0.04 points. The GenAI excess disappears when each venue receives equal weight. Equal venue weighting gives sparse cells much of the weight. The primary estimate and six venue omissions appear in Appendix D.2. Technology is classified at paper level; a mechanism is not assigned to an individual AI condition.

Multiple control categories carry much of the difference. At least two control categories occur in 215/1,069 GenAI-classified papers (20.1%) and 129/1,369 traditional AI/ML papers (9.4%). This counts controls that a paper describes together, not controls that people used together. The multiple-category state accounts for +10.9 of the +13.7-point adjusted family gap on this joint denominator. Standardizing both groups to the same pooled number of non-control categories leaves 19.7% versus 9.6% with at least two control categories (+10.1 points, all 2,438 jointly coded papers). Thus, simply recording more other categories does not account for the pattern. Control and guidance often co-occur in the same paper; learning requires the study comparisons below. Appendix D gives the annual family profiles (Figure 56).

Collaboration and perspective became more common over time. Holding the technology mix fixed at the original analysis's reference composition (43.75% GenAI), its average annual change was +1.10 percentage points (95% CI [0.52, 1.68], Holm $p = 0.005$). The 2023–2025 sensitivity also showed an increase (+2.67 points per year, CI [1.20, 4.14], Holm $p = 0.007$). Guidance and feedback increased in the primary window (+1.58 points per year, CI [0.70, 2.47], Holm $p = 0.008$), but its recent interval included zero ([−2.73, 6.47]). None of the five tests of change in the GenAI–traditional AI/ML difference survived Holm correction in either window. These temporal estimates (Figure 10) answer a different question from the overall technology-group differences in Figure 9; Appendix D.3 reports both temporal estimands for every family.

From task intent to action.

What does the interface make possible?

Five design families · eighteen categories · 110 retained distinctions

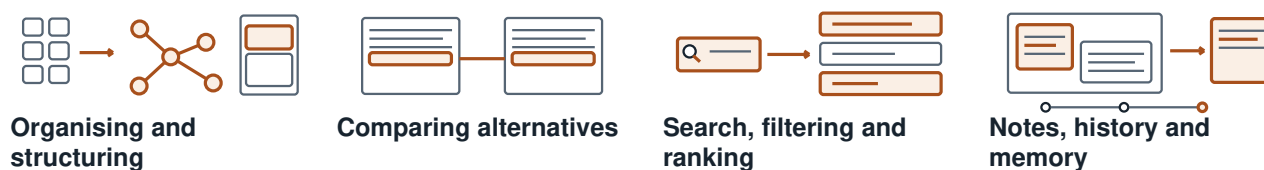
Explanation and visibility

What can a person inspect and understand?



Organisation and comparison

How can a person organise and compare material?



User control and expression

What can a person express, change or try?



Guidance and feedback

How is the next action or reflection supported?



Collaboration and perspective

How can people share work and perspectives?



Figure 8: Five design families organise eighteen categories of semantic scaffolding. Each small interface schematic illustrates a category, with its operation highlighted in orange. A mechanism can combine operations across these branches; layout does not encode prevalence or effectiveness. Complete category maps and source-paper illustrations appear in Appendix C.1. Explore the design atlas; search all 110 distinctions.

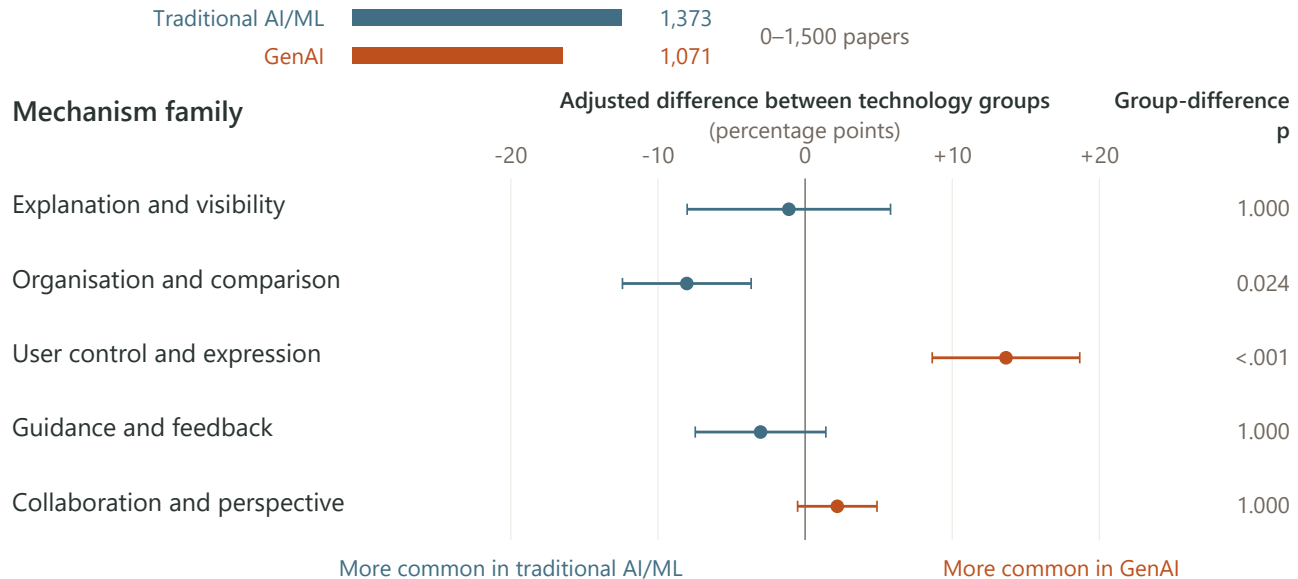
Mechanisms can also be grouped by the activity they support: *specify intent*, *represent possibilities*, *evaluate alternatives*, and *act*

and revise (Figure 11). Appendix C.1 maps all 110 facet codes to these activities; a code can serve several. Annotations, histories,

GenAI papers more often offer user control

GenAI versus traditional AI/ML, accounting for publication year and venue

2,444 papers with family coding and classified technology · 2019–2025



Lines: 95% confidence intervals · Group-difference p-values: Holm-adjusted

Figure 9: Between-technology differences in mechanism families among 2,444 papers with usable family coding and classified technology, 2019–2025. Dots show GenAI minus traditional AI/ML prevalence, adjusted for year and venue; lines show pointwise 95% confidence intervals. The p -values test these group differences, with Holm correction across 51 contrasts, rather than change over time. User control retains support in the 2023–2025 and partial-2026 sensitivity windows. Of 3,385 family-coded papers, 540 lack classified technology and 401 classified papers fall outside 2019–2025.

shared views, and handoffs connect the activities across time and people. The functions describe dependencies, not a fixed order: comparing needs alternatives and criteria, and a comparison may prompt a change to either.

Different mechanisms can serve the same function. Intent elicitation, constraints, and shared criteria frame a task. Structured alternatives and source links make possibilities inspectable; comparisons, explanations, and reflection support judgment; selection, editing, and changed constraints enable action. One mechanism can serve two functions: a comparison exposes options and supports choice, and a constraint can frame a task or redirect it. Guidance can enter at any point, and histories keep context across transitions.

Luminate’s dimensions organize alternatives for exploration and evaluation, with return paths to earlier prompt-specific spaces [119]. AI Chains instead feeds one step’s output into the next step’s input, so people can inspect, edit, and re-run intermediate steps [135]. The order in which a system computes need not dictate the order in which people inspect or repair its work.

An ordered workflow can still require return paths. ColabCoder opens shared comparison after independent coding, then supports discussion, decisions, and grouping while letting people

restore their original codes [41]. Crystalline instead builds a route back into the material: a reminder marks related evidence on pages the developer has visited but not yet attended to, and three of twelve participants reported using it to guide their research [72]. ColabCoder sets an order for the work; Crystalline returns people to an earlier step when their comparison may have a gap. Transitions therefore deserve the same design care as individual mechanisms: keep the criterion that shaped an alternative, the evidence behind a draft, and a route back to an earlier decision, and test whether people understand a transition, change it, and recover its context.

5.3 Revision, preservation, and control

An editable step does not guarantee that people can correct it and keep the later work they still want. Across the cases, the design question is what a human action authorizes the system to reuse or treat as accepted. Table 1 distinguishes saving an old version, preserving valid edits during regeneration, and confirming a result.

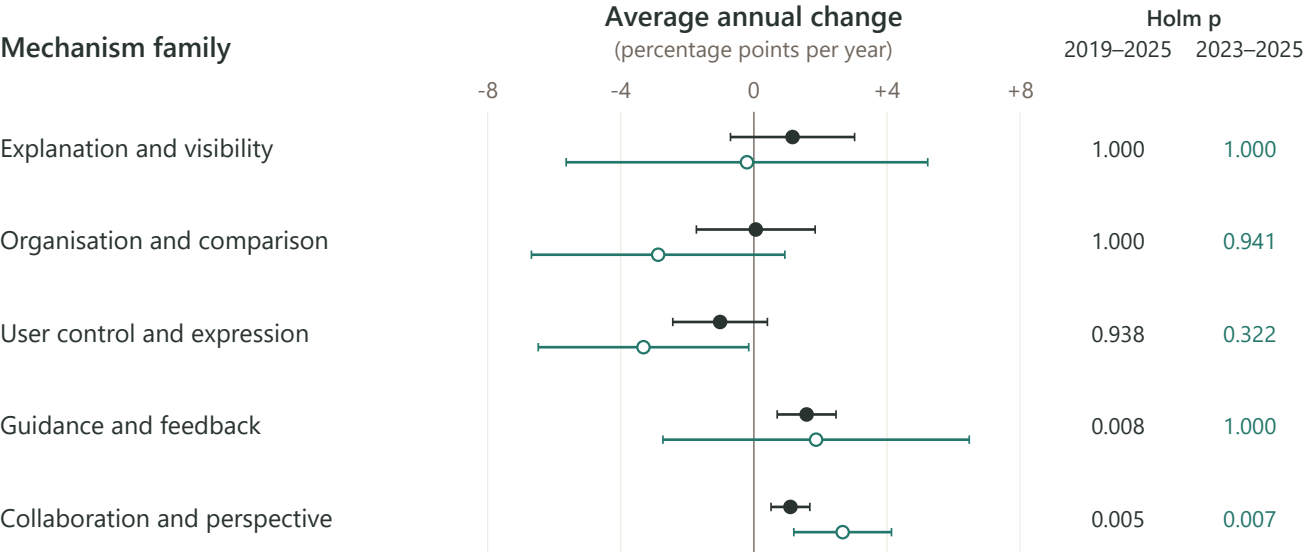
Keeping an old version does not carry its edits forward. Kazemitabaar et al.’s data-analysis systems saved earlier versions in branches, but regeneration ignored later manual edits [57]. Participants described avoiding revisions and expecting to redo or reread

Collaboration and perspective increase across years

Annual change in coded prevalence, with a fixed mix of AI technologies
Adjusted for venue · 39 venues · Partial 2026 excluded

● 2019–2025 · 2,444 papers

○ 2023–2025 · 1,390 papers



Lines: 95% confidence intervals · Holm correction across ten temporal tests per window

Figure 10: Average annual change in coded mechanism-family prevalence: 2019–2025 (2,444 papers) and the 2023–2025 sensitivity (1,390). Dots show percentage points per year; lines show pointwise 95% venue-clustered confidence intervals. The technology mix is held fixed at 43.75% GenAI in both windows. Positive values indicate increasing prevalence. These are temporal slopes, not the overall technology-group differences in Figure 9. Appendix D.3 gives the methods, changing-gap estimates and all ten tests used for Holm correction within each window.

Table 1: Revision and downstream effects. What three interfaces preserve, update, or treat as confirmed.

Design choice	Comparison and finding	Conditions and contrary evidence	Design implication
What survives regeneration?	In task decomposition, editing an assumption, plan or code component can regenerate later work while omitting manual edits [57].	The earlier version remains recoverable in a branch; preservation within the revised result is a different property.	Make the fate of earlier edits inspectable when an upstream decision changes.
What does locking text preserve?	Plume retains locked or manually edited dashboard text while using it to inform further generation [71].	An unchanged passage can still influence other text; locking does not isolate its downstream effects.	Distinguish preserving wording from restricting its influence on new content.
What counts as confirmation?	LineageD permits corrections before predicting the next level; advancing implicitly confirms remaining pairings [46].	Explicit and implicit confirmations share the green marking, so the display does not distinguish how a pairing was accepted.	Preserve whether an item was checked explicitly when that distinction matters to later work.

work; one wanted invalidated parts highlighted instead of regenerated, and another valued the interconnected updates. Recovering an old version differs from preserving the still-valid parts of that work in a new one.

LineageD’s pilot showed the cost of predicting a whole lineage at once: correcting one pairing invalidated the rest of the path. The redesign predicts one level at a time, but moving to the next level implicitly confirms pairings the person has not individually confirmed [46]. Plume keeps locked or manually edited text and

From task intent to action

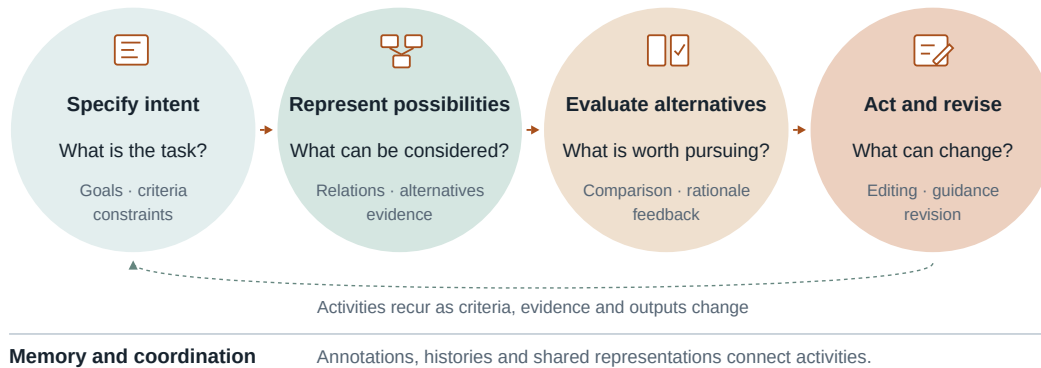


Figure 11: From task intent to action. Four overlapping functions offer an activity-oriented view of the mechanism taxonomy. Memory and coordination connect them across time and people. Arrows show possible dependencies and return paths, not measured transition frequencies. The accompanying source examples illustrate these relationships [41, 72, 119, 135].

uses it as context for later generation [71], so a lock preserves wording without limiting its influence on other text or vouching for its correctness. Relating a local step to the whole process is hard in itself: in AI Chains, four of twenty participants asked how a local change would affect the final result [135].

Interfaces promise to keep different things. Automatic propagation can itself help. CoPrompt participants valued automatic coordination, and eight used history and difference views to resolve conflicts or restore versions [34]. Pail’s code differences, summaries, and design panel supported inspection, but participants reported difficulty following cascades across them, less so when they paused to read the changes [139]. Selecting part of an artifact does not by itself fix what stays unchanged: FusAIn leaves unselected image regions untouched, whereas Textoshop can change text beyond the selected passage [83, 100]. Blace keeps earlier states for rollback, and DesignManager records the process behind an asset [75, 138]; both support recovery and inspection of past work, while consistency with a revised goal still requires checking the downstream artifacts. A design should name the property it protects and the change that could violate it, and an evaluation should test whether people anticipate that change, detect a violation, and recover the work they want to keep.

Edits can change the specification, the model, or the output. A correction can change what the person asked for, how the system understood it, or what it produced. FusAIn lets designers compose and revise a visual specification [100], and Falx synthesizes candidate charts from examples, which participants described checking against input values they knew [126]. CoAutoML lets novices inspect and edit the task, inputs, target, and evaluation assumptions it proposes before they confirm a machine-learning task [118]. Plume’s locks retain part of what the system produced [71]. Outside generative chat interfaces, corrections can also update the model: SmartEye uses a corrected crop as a positive training example for later recommendations [79], and DeepSI fine-tunes BERT from spatial feedback and recomputes its document map,

though its authors note limited visibility into the model’s internal state [13]. The design question is what a correction changes and which later operation uses it, not whether every control comes before generation.

Generating alternatives and their criteria.

When a system generates the criteria as well as the alternatives, people must judge both. In Luminate, selecting an axis rearranges the responses, whereas adding a dimension revises the responses themselves [119], so the writer in our opening example judges the dimensions as well as the ideas. VIVID generates expected answers and rubrics along with candidate dialogues, and instructors inspect and edit them [28]. Design-rationale research already links questions, options, and criteria [81], and conventional tools sort alternatives into categories that designers choose. Generated criteria raise a narrower question: can people revise a criterion while holding the alternatives fixed, or does the revision also replace what they are judging?

Generated criteria can improve comparison. Selenite gives readers generated overviews of criteria and labels the relevant passages; in a twelve-person paired comparison with unaided reading, participants produced more valid criteria with the overview (12.93 versus 6.79), with higher precision and recall [73]. A criterion can also be easy to act on and still miss the quality it stands for. RECAST lets people revise text against a toxicity model; revisions lowered the model’s toxicity scores without reliably improving human ratings of toxicity [133]. When both the model’s response and human judgment matter, the evaluation has to measure both.

5.4 Comparing explanation and practice designs

Explanation content, required reflection, dialogue, and optional recommendations change different parts of practice. Table 2 links four AI-involved comparisons to their assessed outcomes.

Prior learning research separates creating an explanation from using it. In a randomized study of 120 learners, everyone wrote subgoal labels; those who then practiced with their own labels

Table 2: Selected AI-involved comparisons of explanation and practice designs.

Comparison	Assessed result	Conditions
Predicted-misconception explanations [21]	Better immediate exercise-prescription decisions after AI removal than unilateral explanations or no AI	Advantage over random foils unresolved; no delayed test
Required reflection after AI feedback [7]	2.0 versus 3.4 problems completed in eight minutes; transfer and code-evaluation advantages unestablished	Fixed time budget; fewer attempts do not establish learning harm
QuizBot versus flashcards [107]	Higher recognition (87.2% versus 65.8%), with an advantage at two months	Equal repetitions; 46 versus 14.3 minutes of practice; dialogue bundled with other support
Adaptive versus generic video recommendations [66]	Small algebra-ability advantage during normal schooling	Low optional uptake; policy effect; closure-period advantage unresolved

scored 21.6 of 25 on an immediate test, against 17.1 with expert labels and 18.8 with none. Instructional materials were removed for the test, while the programming environment remained available [82]. This comparison sharpens ExPeerience’s question (Section 4.2): which part of the task should the person learn to perform? Learning research has long stressed what support asks learners to do [27, 104]; the interface choice is what to generate and what to leave for the learner.

Writing feedback can describe a revision or judge its quality. ArgRewrite classifies the purposes of a writer’s earlier revisions without judging their quality, and asks writers to agree or disagree with those labels before and after revising again [1]. ArgueTutor pairs claim and premise labels with recommendations based on their balance [125], and LegalWriter adds advice on argument components while leaving the semantic fit between claim and conclusion to the writer [129]; both improved the quality of text written with them over their comparison interfaces. A description of a revision and advice about quality make different assertions, and one interface can offer both. A learning claim needs a test of the judgment or revision skill the feedback is meant to build.

5.5 How prior knowledge informs design

Prior knowledge enters a design through the material it supplies, the activity it asks of a person, and the measures used to assess it. The same learning principle can motivate different choices at each point. We traced how 2,058 papers, including the non-AI comparison papers, used the prior work they cite (51,815 source records; a work cited by two papers counts twice), and we compare implementations through the passages that document each use.

Tracing sources into designs. The citation map (Figure 13) ranks the sources used by the most papers and follows the highest-ranked source with a known year into the later designs that use it; each source–paper pair counts once. The companion website lets readers search sources, follow connections, and open the passage behind each use.

Making correspondence inspectable. Kori turns the concern about split attention into a visible link between prose and chart marks [67]. The system proposes links, and authors preview, accept, discard, or draw them by hand; once a link is accepted, hovering over its phrase highlights the chart marks it refers to (Figure 12). The author checks each correspondence, and the reader uses it to interpret the data.

Operationalizing a theory is not testing learning. QuizBot cites ICAP [27] as its learning rationale, whereas the Khanmigo

methodology turns ICAP into an LLM-judged indicator of cognitive engagement [107, 122]. Deeper use of the theory does not bring stronger evidence of learning: QuizBot tests recognition two months later, while Khanmigo tracks correctness on the next item a student attempts alone. A theory can say what to observe without showing that changing that observation produces learning.

Prior knowledge can shape the evaluation as well as the design. Sibyl adopts Doshi-Velez and Kim’s application-grounded evaluation [33, 148], testing explanations with screeners in child-welfare work. Screeners often rated case-specific explanations helpful and found other forms less useful during those cases, and their feedback exposed problems of terminology and presentation.

Supplied material can support learning through its use. Learners need not build the material they learn with. VocabEncounter supplies phrases in context and reports better vocabulary two days later than its word-only condition [5]. Rehearsal supplies conversational alternatives, practice, and feedback; its participants used more cooperative strategies in a later unaided conversation than participants given the same static materials without interactive practice [111]. In a study of AI assistance for incidental learning, explanations without a recommended answer improved immediate concept learning over minimal feedback [39]. Supplied material also leaves room for different uses: CoNotate participants reported both running its suggested queries and borrowing their vocabulary to write new ones [95]. The design question is which operation the supplied material helps people perform, and whether the study tests that ability.

5.6 What capability the evaluation establishes

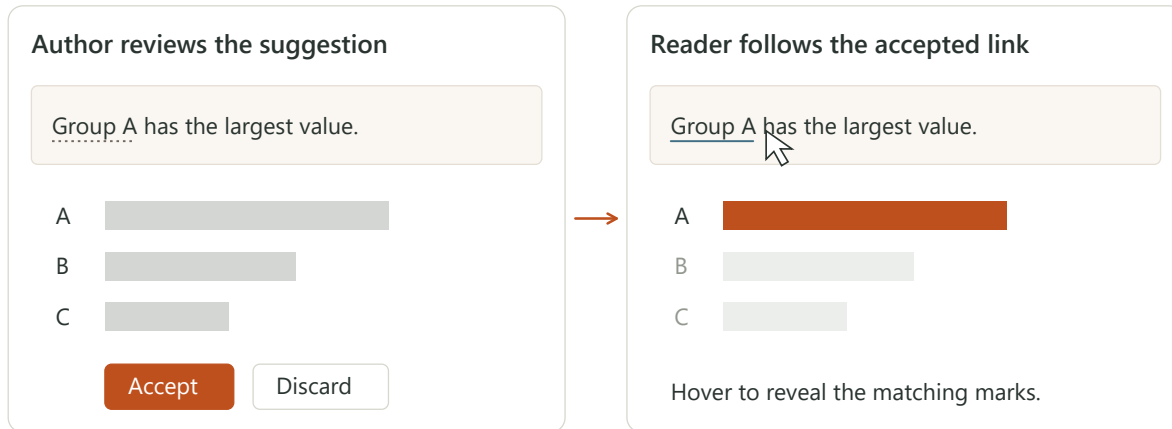
We distinguish process, artifact, and capability outcomes, assessed by the study or reported by participants. Each measure specifies its construct, scoring criteria, rater, comparison, timing, and available assistance.

Product and capability measures answer different questions. In BELVEDERE, the maps showed more explicit links between evidence and hypotheses, but no difference in final prose conclusions was detected between conditions, and students could still consult their maps while writing [120]. Both outcomes describe products made with support; neither tests what students can do without the maps.

A benefit depends on the comparison and the situation. Cognitive forcing asks people to engage more deliberately with AI advice. Among 199 participants, fully correct decisions averaged

Kori makes text–chart links inspectable

AI proposes a connection. The author decides whether to keep it.



The author checks the correspondence; the reader uses it to navigate the data.

Figure 12: From an AI suggestion to an interactive reference in Kori. The author reviews a proposed text–chart connection. After acceptance, hovering over the linked phrase highlights the matching marks by fading the others [67]. Schematic redrawn from the paper’s interface description, with illustrative data.

35% with simple explanations and 33% with forcing, a nonsignificant difference; on items with incorrect AI advice, forcing raised correctness from 3% to 9%, while the no-AI condition reached 18% [20]. Adaptive help raises the same issue: predicting whether a person will answer correctly differs from showing that advice based on that prediction helps them [77, 117]. Appendix A.7 gives the learning and duration profiles of the reviewed studies.

Independence concerns particular people, tasks, and resources. Useful tools need not be withdrawn. Pea distinguishes developing a person’s capability from removing resources that stay useful [99]. In a study of vocational prompting, six of eight participants reached full task correctness with a prompting device, and asking a person for help lowered the score while consulting the device did not, so the score measured independence from a helper, not unaided performance [25]. A design should name which activity is meant to become independent, which resources should remain, and who prepares or maintains them.

5.7 Performance and subsequent capability

An interface can improve one assessed outcome while leaving another unresolved. Table 3 distinguishes comprehension, revision quality, enacted behavior, and application of supported skills.

These comparisons apply Norman’s distinction between performance with an artifact and the capability attributed to the person [91]. DBox’s decomposition and feedback also improved performance on an immediate similar problem solved without assistance, although effort rose [78].

Assessment conditions specify the capability demonstrated. Among the 396 AI-involved sources in the outcome appraisal, 223 reported a separate capability assessment, 31 assessed capability in a later session, 71 used new material or a changed task, and 67 reduced or removed a named training aid. These overlapping features describe the selected appraisal set (Appendix A.8).

An assessment’s task, timing, and available support each change what it shows: an immediate new problem can test application without testing retention, and a later test can still allow help. WriteAid tested pupils’ writing two to three weeks later without AI or teacher support, but on the same topic and without an untreated comparison or a pretreatment skill measure [90].

The promised use determines the informative test. Immediate performance, later retention, and application to new material need not move together. Two classroom experiments with structured LLM reflection found no significant advantage on exams two weeks later [64]. Preference is not capability either: across three AI Margin Notes studies of reader interfaces, note selection, and human and AI note-taking, comprehension two hours after reading did not differ significantly, even where preferences or task demands did [55].

An informative evaluation therefore starts from the design’s intended use. An explanation meant to prepare a new decision needs a decision task; feedback meant to change the next attempt needs that attempt observed; a practice partner meant to build interpersonal skill needs evidence of enacted behavior, while recall answers

Table 3: Assessed outcomes in four AI-involved comparisons.

Comparison	Assessed result	Conditions
CodeVoyager versus same-model chat [59]	Higher immediate closed-book comprehension (8.06 versus 5.44/20); no detected assisted-completion or accuracy advantage	16 developers; interface package; no delayed test
FeedbackWriter versus usual TA feedback [74]	Better revisions ($d = 0.5$); knowledge-test advantage unestablished	Randomized 354-student trial; optional posttest; test-time assistance unclear
Rehearsal added to static training [111]	Better immediate conflict-response behavior; recall and recognition advantages unestablished	Behavior and recall assess different capabilities
CodeTailor versus generated solutions [49]	Better immediate unsupported application of supported elements; total posttest advantage unestablished	Selected 14 of 18 participants; tested elements depend on practice difficulty

a different question. Effectiveness reviews already show that scaffolding can improve cognitive outcomes [11]; the comparisons here show which alternative and which outcome a designer can use as evidence.

6 DISCUSSION

The design space shows what researchers have built. The closer comparisons show that designs sharing a label can ask very different things of people. An edit can repair how the system read a request or change the finished artifact. A shared representation can carry a learner’s priorities to a coach or stand in for a counterpart in practice. We discuss what these differences mean for design.

6.1 Support that stays can still teach

When learning is a goal, design for a named capability as well as for the work produced. Norman separates what a person and an artifact accomplish together from the person’s own activity, and Reiser separates structuring a task from drawing attention to what needs reasoning [91, 104]. Krakauer’s distinction between artifacts that leave capability behind and those whose advantage lasts only while in use asks what people keep when the artifact is gone [63]. The capability can include using a resource well. Pea separates scaffolding meant to fade from support meant to stay [99]. In the vocational-prompting study, a step completed correctly with the device counted differently from a step for which the worker asked a person for help: the target was a specific independence, not doing without every aid [25].

6.2 Choosing what people can inspect and change

When a learner must tell alternatives apart, show the contrast that decides between them. The contrastive-explanation experiment and Incremental XAI vary what to explain, how to organize it, and when to show it; Incremental XAI also tests understanding on new cases after the explanations are removed [16, 21]. These are the choices hidden inside “how much help” in the assistance dilemma [62].

Making the object of correction explicit. Correcting a misread request, stating a preference for later suggestions, and changing the finished artifact are different operations. CoAutoML exposes task assumptions before execution; SmartEye learns from corrected

outputs [79, 118]. People can choose the right intervention only if the interface shows what each edit will change.

Following support across roles. When the learner and the person using the interface differ, the design must carry what the next person needs to help. The coaching systems expose a novice’s priorities or a tutor’s activity for another person’s judgment [17, 51]. The learner’s contribution should stay traceable, and open to correction, across that handoff.

6.3 Judging generated criteria

Generative systems can supply the criteria as well as the alternatives. Luminate proposes dimensions for comparing ideas, and VIVID proposes rubrics alongside content [28, 119]. Questions, options, and criteria are old material for design rationale [81]; these systems make them generated and revisable during use. That adds a task the writer in our opening example faces: judging a proposed criterion before judging anything against it.

6.4 Budgeting the activity support adds

A required activity spends practice time. Reflection after AI feedback cut completed practice in a fixed interval, and QuizBot’s fixed number of repetitions let time demands differ between conditions [7, 107]. The two answer different questions about how to spend a budget. Design an added activity around the skill it should build and the practice it displaces. Assigned support also differs from used support: the algebra-video trial compares recommendation policies, not watching a video [66], and ONYX and Leap make references available while uptake and successful correction remain separate outcomes [35, 108]. For a checking task, name the uncertainty, the reference that resolves it, and the correction the person can make.

6.5 Assessing the capability you claim

When the goal reaches beyond the current task, design the assessment around the future activity. CodeVoyager’s closed-book comprehension result and its assisted completion measures answer different questions, and FeedbackWriter improved revisions without an advantage on its optional knowledge test [59, 74]. To test acting without an aid, remove the aid in the assessment. To test reasoning with a resource that stays, test new cases with the resource available.

7 RESEARCH OPPORTUNITIES

Each study below names what changes in the design, whose activity to observe, and what would support a learning claim.

7.1 Making corrections take effect

Let people repair the same mistaken constraint either in an intermediate representation of the task or in the resulting artifact, as CoAutoML, SmartEye, and Plume variously allow [71, 79, 118]. Hold the model, the initial error, and the evidence constant. Record whether the correction survives the next transformation, what unrelated content changes, and whether people catch remaining errors; a later task with a new error tests whether people learned the dependency.

Preserving a choice when its basis changes. ABScribe, Notable, and Plume preserve different objects of revision [70, 71, 105]. Compare preservation alone with preservation plus a cue that a premise changed. Give each condition one invalidated choice and one still-valid choice, and measure error detection, unnecessary revisions, and final correctness. In creative work, record the intended constraints so that a deliberate reinterpretation can be told apart from a failed correction.

Correcting an earlier step after later edits. Green and Petre’s cognitive dimensions name the concerns at stake: viscosity, the effort a change takes; hidden dependencies; and progressive evaluation, checking work before it is complete [43]. Human–AI guidelines also ask systems to show the consequences of a person’s actions [4]. The revision cases suggest a test: after a person edits a later result, ask them to correct an earlier step, then check which still-valid edits survive, whether the person finds the affected outputs, and how much work and how many errors rechecking takes. Selective reapplication, regeneration, and restoring an earlier version may each support a different part of this task.

7.2 Tracing support through people

Give coaches either a summary alone or the same summary with the learner’s priorities and traceable contributions [17, 51], with equal preparation time. Follow what the coach infers, what the learner corrects, and which issues reach the next meeting, and analyze coach–learner groups together. A generated practice counterpart plays a different role [111]: its replies are practice material, not evidence of how a real person would respond.

7.3 Isolating what supplied material contributes

For the same decision task, compare explanations that name no alternative with explanations contrasted against a predicted misconception or a random alternative. Compare schedules that present the same factors together or one at a time [16, 21]. Test immediate and delayed decisions on new cases without the explanation display, and estimate the planned contrasts directly. For generated criteria such as Luminate’s dimensions and VIVID’s rubrics [28, 119], compare ways of supplying and revising the same criteria, measure what they omit as well as agreement with them, and use an independent measure of task quality.

7.4 Comparing required activity and actual use

Compare reflection after feedback with feedback alone in one programming task, under equal time and under equal repetitions, with the same item bank and feedback [7, 107]; test near and far transfer immediately and after a delay. For optional help, compare adaptive and generic recommendations with the same videos and quiz schedule, and report effects of the assigned policy separately from associations with voluntary viewing [66]. For checking, compare an inspectable reference with the same reference plus a targeted prompt, include correct and incorrect proposals, and test a later changed case with the prompt removed [35, 108].

7.5 Assessing learning with continuing resources

Keep the actual alternatives, such as visualization versus chat on the same model or AI-assisted versus usual teaching-assistant feedback, and test new material with a common assessment [59, 74]. Name what help remains during the test: the subgoal-label study kept the programming environment, and the vocational study removed immediate human help [25, 82]. Stating the task, comparator, timing, available help, and participation in advance turns each proposal into a test of the capability claimed.

8 LIMITATIONS

Coverage. We searched 40 venues over 8 years, which favors published, English-language research [6, 69]. Technology comparisons leave out 639 papers whose AI technology could not be determined, and each analysis uses the papers coded for its fields (Table 8). The theories and empirical studies we compare against are a selection, not a census of every relevant tradition.

Reading and coding. Language models read each paper against a fixed codebook, one model per paper, and grouped the coded descriptions into categories. People have not yet checked the coding or the cross-case interpretations. Models can miss evidence shown only in figures and can share biases, and other analysts could draw the boundaries between closely related mechanisms differently.

Effects. Studies differ in tasks, participants, comparators, outcomes, and duration, and many compare whole systems. Most coded changes in how people work rest on the authors’ own attribution rather than a measured contrast (75.8% of coded process shifts). The GenAI and traditional AI/ML comparisons are associations among published designs, not effects of the technology. The design space cannot rank mechanisms by effectiveness or estimate an overall learning benefit.

9 CONCLUSION

AI tools increasingly supply the structures people think with: criteria, examples, plans, and records of earlier choices. Our design space maps these semantic scaffolds across domains and connects each to the work it organizes and to what studies found. Generative-AI papers more often concern producing artifacts and give people more control over the system. What people can change matters, whether a specification, the AI’s interpretation, the criteria, or the result, and support can reach a learner through peers, teachers, and coaches. Supplied material can support learning through the way

people use it, and added interaction does not by itself make people practice the intended skill. The design space gives researchers concrete alternatives for building help that develops capability as people work, including the capability to use support that stays.

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A EVIDENCE EXTRACTION AND CODING

This appendix holds what a reader needs to check or reuse the coding: the fields of each evidence card, how the closed and open fields were coded, the agreement checks, the data behind each analysis, how sources for the outcome comparisons were chosen, and how cited sources were matched.

A.1 Card Fields

Each Stream 1 paper has a mechanism card (Table 4), and 2,058 papers also have a theory-lineage card (Table 5); each Stream 2 source has a knowledge-claim card (Table 6). The sectioned layout follows Model Cards [87], and AI-assisted design-card generation is a precedent for turning source findings into cards [115].

Table 4: Stream 1 mechanism card: work and human activity first, then scaffold design and evaluation.

Card section	Fields recorded
Workflow changes (RQ1)	Process before and after support; mechanism credited; attribution basis; source of the baseline.
Human activity (RQ1)	Activity externalized to the tool or representation; activity retained by people; work still required.
Scaffold design (RQ2)	Author term; facet and value, with their merged codes; what makes each mechanism separate; supporting passage and page.
Learning and support (RQ2)	Learning measurement, direction, and evidence; reliance; changes in support and their grounds; observation horizon.
Measures and comparisons (RQ2)	Recorded outcome layer (process, artifact, learning); author claim, claim scope, contribution type, study design, participants, comparison, statistics, and claim–evidence alignment.
Conditions and limitations	Author-stated limitations and boundary conditions.
Recorded codes	Work domain, intent, expertise, capability gap, support properties, AI involvement, and technology era.
Source	Paper ID, DOI, venue, year, page count, the PDF read, the full-text decision, a map of relevant pages, and codebook notes.

Table 5: Stream 1 theory-lineage card: how the paper uses the sources it cites (RQ2).

Card section	Fields recorded
0. Identity and position	Link to the mechanism card; overall position toward prior work; position summary; human-facing rationale; lineage depth.
1. Documented sources (RQ2)	One entry per cited work: citation identity, relation to the focal paper, use, enactment, and supporting passage and page.
2. What was read	Introduction and related-work pages; design or evaluation pages checked for application of a cited idea.
3. Synthesis note	A short account of what the paper’s use of prior work shows.

Table 6: Stream 2 card: one to eight field-level claims, linked to the research questions they inform.

Card section	Fields recorded
0. Identity and position	Card and work IDs; publication details; discovery routes; contribution types; position summary; overall uncertainty.
1. Page-grounded claims	One entry per claim: author wording and reviewer interpretation; question links; applicability; affected human work; outcome evidence; evidence basis; counterevidence and limits; supporting excerpts; uncertainty.
2. Evidence store	Verbatim excerpts with pages and locators; links recording which claim each excerpt supports, its role, and the reviewer paraphrase.

A.2 Closed Fields

Closed fields take values from a fixed list. Intent records whether the support targets immediate performance or sensemaking, learning or skill development, or both. Measurement records what the evaluation assessed, so an intention to support learning stays apart from a

Table 7: Blind open-code comparison on exactly matched observations. Codes gives existing / blind code counts; Rand is raw pair agreement, and pair F1 compares the pairs grouped together. Unmatched counts existing observations outside the comparison. Papers overlap across fields, so the 1,571 matched observations are not 1,571 distinct papers.

Field	Obs. (papers)	Unmatched	Codes	Rand	ARI	Pair F1
Externalized to tool	110 (110)	18	64 / 46	96.2%	.167	.186
Facet	198 (48)	15	132 / 76	98.0%	.132	.142
Process before	125 (48)	15	79 / 34	95.8%	.445	.462
Process after	419 (160)	0	262 / 60	97.0%	.079	.088
Value	695 (144)	0	517 / 95	98.1%	.082	.087
Learning evidence	24 (24)	24	5 / 16	50.4%	.108	.217

measure of it; the direction of a result and its observation horizon are recorded with the measure. Support properties are coded separately and can co-occur: structuring, problematizing, contingency, adaptation, fading, responsibility transfer, and distributed support. Contingency responds to the person’s activity or need. Fading reduces support as competence grows; a fixed reduction on a schedule is not by itself fading. Responsibility transfer concerns who performs or regulates the activity, not whether the interface remains available. Technology is classified once per paper, from the AI in its system, an experimental condition, training, or evaluation, so a paper classified as generative AI can still contain mechanisms that involve no generation.

A.3 Open Fields

The 21 open fields hold 142,877 observations in the papers’ own terms. Equivalent descriptions within a field were merged into 1,290 codes, and the codes were grouped into 223 categories and 92 families across 18 fields, with subcategories only where they add a distinction. Each grouping keeps links to its observations, and fields are never pooled. Codes that record missing information, such as an unspecified baseline or an unreported measure, stay apart from substantive categories. When one free-text value lists several limitations, its single code summarizes them rather than listing each. Appendix B defines the families and categories of the six fields the framework uses.

A.4 Agreement Checks

To test the open-code grouping, a separate model pass coded extracted observations with the existing codes hidden, and we compared the two groupings field by field (Table 7). An observation entered the comparison only when its identifier, source card, and field value matched exactly. Two codings need not use the same words, so we compared partitions: whether each pair of observations shares a code in each coding. The adjusted Rand index (ARI) corrects pair agreement for chance, and positive-pair F1 summarizes overlap among the pairs either coding groups together; neither treats one coding as correct.

Raw agreement is high because both codings keep most pairs apart: for facet, 19,083 of 19,503 pairs are apart in both and 32 are together in both. The blind pass used fewer codes than the existing coding in five fields and more in learning evidence. It reported medium or high uncertainty on most assignments in five fields (for example, 659 of 695 values) and on 7 of 24 in learning evidence. The learning-evidence row covers 24 of 48 existing observations; the other 24 share a repeated not-measured value and fell outside the comparison’s selection rule.

A separate check asked only whether AI was involved: three fresh model sessions labeled 532 title-and-abstract records added in September and gave the same label to 529 (99.4%; $\kappa = .988$).

A.5 Validation Status of the Readings

For part of the evidence, the records of the original reading runs have not been fully validated, and we mark that evidence for provisional use. On 26 September 2026, such evidence appeared in 319 of the 1,076 learning papers and 1,252 of the 4,086 papers behind the main figures. The counts mark incomplete validation records, not errors, and the marked papers stay in the analyses.

A.6 Data Used for Each Analysis

Each analysis uses the papers coded for the fields it needs (Table 8).

All 2,847 open-coded papers have complete domain and retained-work codes; 532 lack complete facet–value codes. The closed-field population includes 44 papers outside the open-field profile and two with unresolved support-property coding.

A.7 Learning Measurement

Among the 2,891 closed-field papers, 301 (10.4%) measured learning. Two-thirds of those (199, 66.1%) measured it within a single session; 35 used repeated sessions, 38 followed people over weeks, and 25 measured learning in deployed use.

Table 8: Data behind each analysis. Counts refer to the stated unit.

Data	Count and unit	Analysis
Stream 1 corpus	4,511 papers on 26 September 2026; 474 non-AI comparison papers outside the systematic counts	Corpus counts (Section 3)
Main figures	4,086 papers (4,115 records) with usable coding	Annual, technology, and mechanism figures
Closed fields	2,891 papers coded by 28 August 2026; 301 measure learning	Intent, learning-measurement, and process counts
Open fields	142,877 observations in 21 fields from the 3,295 papers coded by 28 August 2026, of which 2,847 remain in the corpus	Design space and framework values; counts of papers carrying a value
Complete mappings	2,315 papers with complete facet–value codes; 2,311 also with remaining-work codes	Mechanism-by-work matrix (Appendix C)
Theory lineage	2,058 papers, including non-AI comparison papers; 51,815 cited-source records	Roles of prior work (Section 5.5)
Stream 2	512 sources; 2,867 claims	Prior findings and theory
Citation network	4,751 papers from both streams	Citation atlas and co-citation map (Section 5.5)

A.8 Selection of Sources for Outcome Comparison

The study comparisons in Section 5 draw on sources chosen for their outcome evidence. A learning-scope assessment of 4,714 records proposed 1,226 within scope. We appraised the 422 of these whose cards recorded that learning was measured (twelve more had been appraised earlier and were analyzed separately), plus 14 performance cases found among 53 descriptions that needed interpretation: 436 records, or 434 distinct PDFs. The 190 sources flagged as having a documented capability comparison entered the detailed synthesis.

A.9 Source Matching and Coverage

A.9.1 Source-use coding. Each theory-lineage record links one cited work to its role in one citing paper’s design, argument, or measure (Section 5.5), so a work cited by two papers yields two records.

A.9.2 Citation metadata. The bibliographic map joins the two streams by DOI or Semantic Scholar paper ID and counts a paper found in both once. It uses Semantic Scholar data retrieved on 9 September 2026 for 4,331 Stream 1 papers and 448 Stream 2 works (28 in both). Reference lists for 3,233 papers name 106,523 cited sources; two with conflicting identities are withheld. Of the remaining 106,521, 5,748 are cited by at least five review papers and 157 by at least 40 (Figure 14). A reference list shows that a paper cites a work; its lineage card shows how the paper uses it.

The co-citation map counts each citing paper once per source and keeps sources with at least five citers and pairs with at least three shared citers. It weights a pair by association strength: the pair’s shared citers times the number of papers, divided by the product of the two sources’ citer counts. Louvain clustering (seed 42) forms the neighbourhoods, and a maximum spanning forest supplies the drawn links; stem, branches, and leaves mark how widely a source is shared. Less-cited sources sit near their most shared neighbourhood without joining it. The interactive atlas supports zooming into a neighbourhood and inspecting each source.²

A.9.3 Matching and unavailable sources. Cited works without a DOI are matched by title and year. Title variants can split one work into several identities, which overcounts distinct works and undercounts shared references; descriptive labels and conflicting identities stay out of the shared-reference map. Papers without a reference list, and full texts we could not obtain or read, are outside the source-use analysis.

B THE FRAMEWORK’S CODEBOOK

The framework combines mechanism design with five dimensions of human work and evidence. Tables 9 and 10 define those five dimensions; Appendix C.1 gives the mechanism categories and examples. Counts use the 28 August 2026 coding populations in Appendix A.6; a paper may carry several values. Each group names the paper denominator for its percentages. Daggers identify absent or unspecified values, as defined in each row.

B.1 What follows for the person, the work and the evidence

²Interactive citation atlas: the interactive citation atlas .

Table 9: The three dimensions about the person: what the person still does, keeps, and hands to the tool, with the share and number of papers carrying each value. A dagger marks the absence of a value.

Dimension and value	Definition	Share	Papers
What the person still does (n=2,842 papers)			
The task the tool did not do	Work of the task itself that the system does not do: performing the substantive task, making sense of what is shown, and checking and repairing what comes back.	51.1%	1,451
New work the tool created	Obligations that exist only because the system exists: feeding and instructing it, and operating and coping with it.	29.9%	851
The whole task kept by the person	The person feeds the system and judges what comes back, across three or more stages, with no stage removed.	10.6%	300
Final authority kept by design	The final decision, the reliance decision, values-laden judgement and accountability stay with the person as a matter of design or law, not as a step in a workflow.	8.3%	237
Duty asserted, nothing specified†	The value says responsibility remains with the person without naming any obligation.	0.1%	3
What the person keeps (n=2,847 papers)			
Judgement and decision	Interpreting and inferring, deciding and authorising, setting direction, bringing knowledge.	48.0%	1,367
Doing and revising the work	Doing the work, repairing and revising, steering the loop, improving the system.	35.7%	1,016
Checking the output	Checking output for errors, calibrating reliance, judging fit to goal, verifying against outside evidence.	9.1%	260
Responsibility, social and self-directed work	Coordinating and deliberating with others, consent and disclosure, ethical judgment, communicating outward, holding accountability, care work; and learning, reflection and sustained attention to the system.	6.9%	197
Not stated†	The paper does not say what the person retains.	0.2%	7
What moves into the tool (n=2,326 papers)			
Computation and judgement	The tool predicts, simulates, fits models, calculates, optimises, estimates uncertainty, computes geometry; and attributes behaviour, classifies, detects errors, critiques, scores, infers a person's state, ranks and verifies.	30.5%	709
Representation and translation	The tool gives external form, converts between formats and languages, lays material out, summarises, verbalises, encodes data visually, coordinates views, compiles specifications; and finds groupings, patterns, similarities, aggregates and comparisons.	23.6%	549
Decision, guidance and action	The tool recommends, selects the next step or decides outright; supplies templates, procedures, decompositions, reference material, instruction and prompts; propagates judgments, does clerical work, operates environments, enforces constraints, fabricates; stands in for people, coordinates them, relays communication.	15.6%	364
Perception, search and memory	The tool senses, extracts, recognises, segments, transcribes and flags; finds and narrows material; records, stores, recalls and reminds; watches and tracks.	15.0%	349
Content generation	The tool generates new artifacts (alternatives, images, prose, code, 3D forms, audio) and responses (explanations, questions, answers, utterances, conversational replies, revisions of the person's material).	14.9%	346
Offloading that failed or was withheld	The offloading failed or was incomplete, or the tool deliberately stopped at a proposal and left the judgment with the person.	0.3%	8
Nothing offloaded, or not stated†	The value states that nothing moved to the tool, or names no operation at all.	0.0%	1

Table 10: The two dimensions about the work and the evidence: what happens to the work and how learning is evidenced, with the share and number of papers carrying each value. A dagger marks the absence of a value.

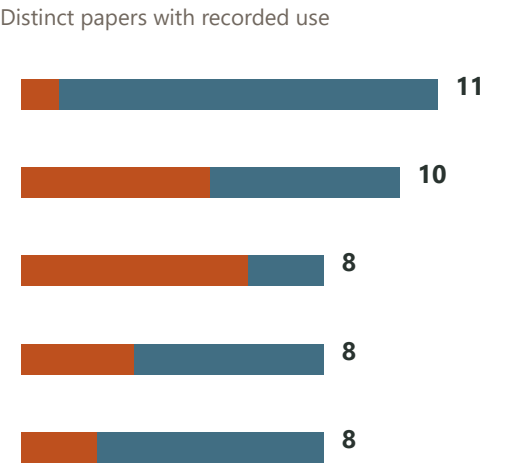
Dimension and value	Definition	Share	Papers
What happens to the work (n=2,643 papers)			
The person's work changes form	The person still works, differently: reading new representations, navigating and searching, steering and specifying at a higher level, reviewing and selecting instead of producing, checking and repairing, following a guided procedure, judging and reflecting on new grounds, coordinating through shared artifacts.	92.2%	2,437
A step becomes automatic	A step the person used to do is now done, or mostly done, by the system: automatic extraction or generation, exception-only attention, the same task made cheaper, automatic checking of the person's own work, effort freed for other work.	26.4%	697
Study condition or proposal, not practice	The value describes a study arm, protocol or manipulation, or a proposed or recommended change that was not observed in use.	16.1%	425
Work newly possible	The person can now do something they could not: work in a new setting or moment, at a larger scale or scope, or a task that was out of reach.	4.6%	121
New work created by the system	Labour that did not exist before the system: setting up, configuring and maintaining it, added steps or obligations, burden or unproductive activity it caused.	4.0%	105
Work unchanged	The practice did not change: the person keeps the core task and the system only informs it, under-uses or works around it, or nothing changed at all.	2.0%	54
How learning is evidenced (n=2,800 papers)			
A proxy measured instead	Capability stood in for by a proxy: subjective judgements about the system, usability instruments, qualitative accounts, attitudes or engagement; in-session task performance, behaviour or rated work products; or the quality of the system's own output, including the machine's learning.	37.6%	1,054
Learning measured	An objective capability measure was taken: immediately with support present, later or with support withdrawn, or with a null, mixed or adverse result, or compromised by confounding or deliberate suppression.	9.4%	263
Learning self-reported only	A learning or skill gain is asserted on self-report alone, or gestured at in remarks while the paper says learning itself was not evaluated.	4.6%	130
Learning not measured†	Nothing about the person's learning, transfer, retention or capability was measured: the value names no measure and no substitute, the authors say so themselves or defer it, there were no participants at all, or learning is the stated purpose yet never evaluated.	48.3%	1,353

Where prior work is used

1,614 distinct papers · 1,669 PDF-matched lineage cards

Most shared identifier-linked sources

- Manifold: A model-agnostic framework for interpretation and diagnosis of machine learning models
- Rulematrix: visualizing and understanding classifiers with rules
- The eyes have it: A task by data type taxonomy for information visualizations
- Kori: Interactive Synthesis of Text and Charts in Data Documents
- NL4DV: A toolkit for generating analytic specifications for data visualization from natural language queries



Follow a source into later designs

Highest-ranked source with an unambiguous recorded year; every use is shown.

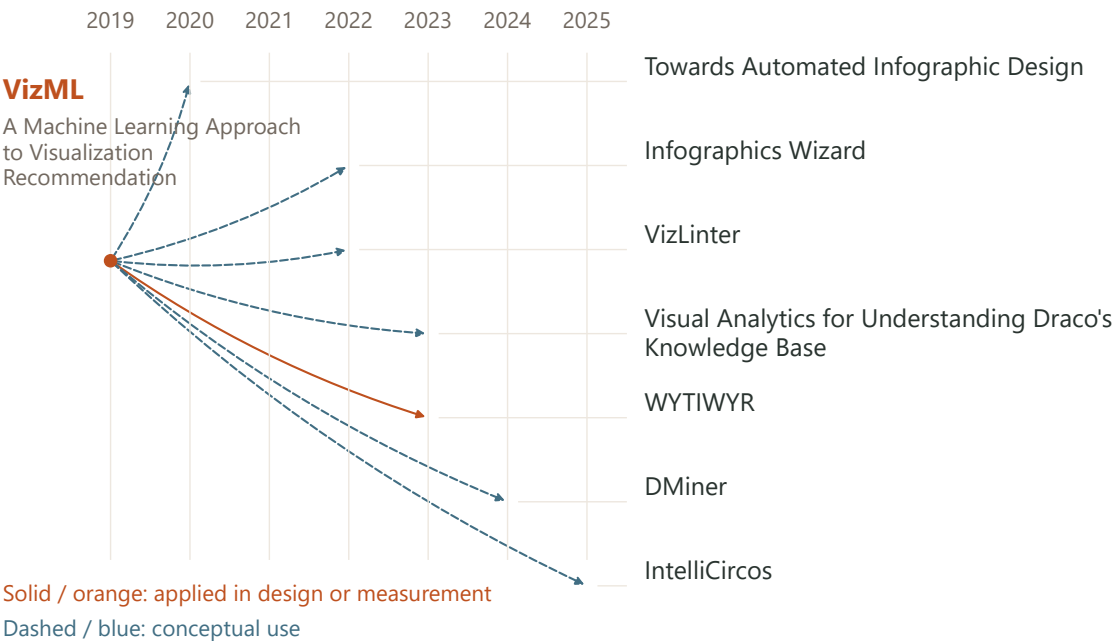


Figure 13: How papers use prior work. Across 1,669 lineage cards for papers in the analysis, the upper panel ranks sources with a DOI or other identifier by the number of papers that use them, split into applied and conceptual-only use. Below, the top-ranked source with a known publication year connects to later designs. Solid arrows show application in design or measurement; dashed arrows show conceptual use. The companion website links all 41,266 source–paper pairs to passages and cards.

Shared sources across the two review streams

106,521 cited sources · 5,748 cited by five or more review papers

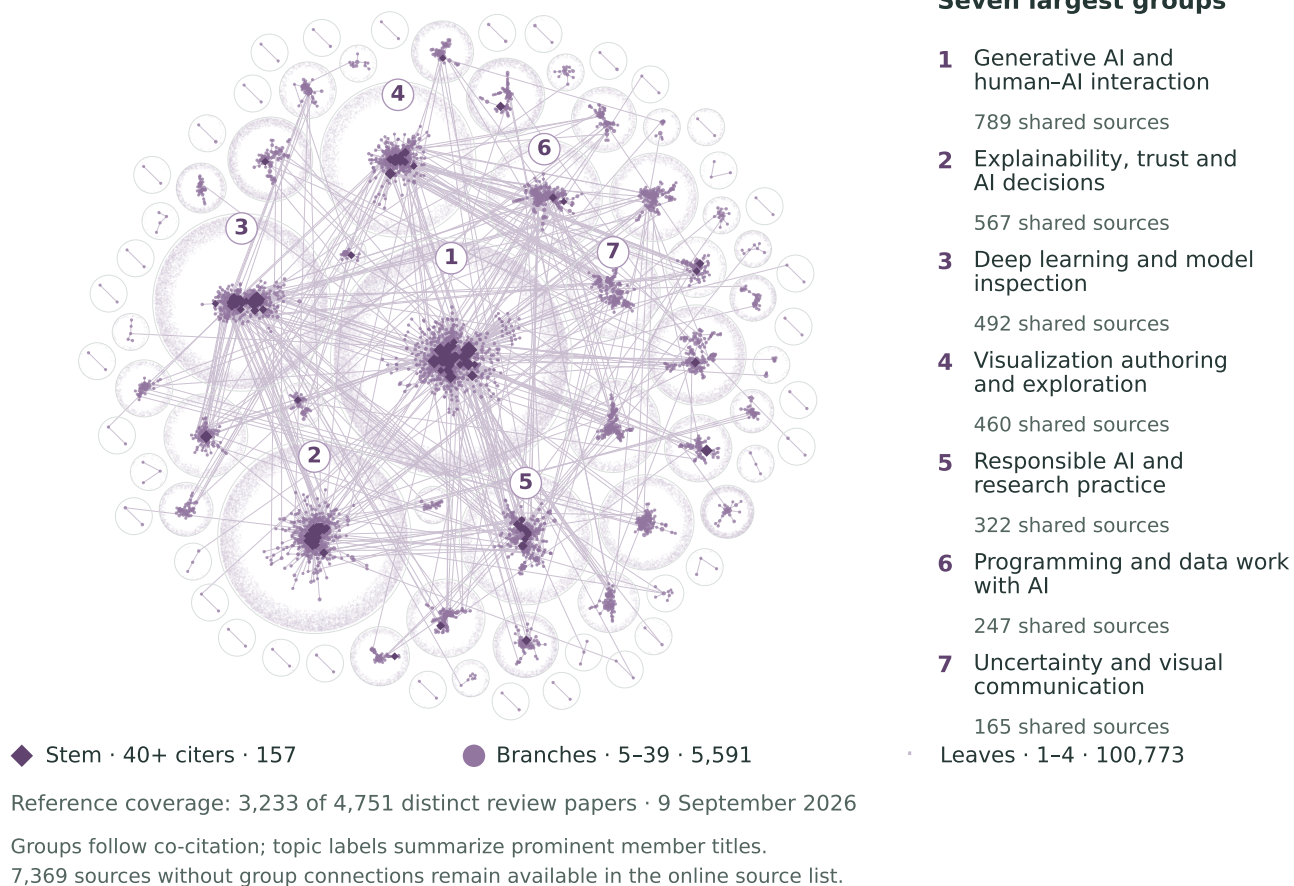


Figure 14: Shared-reference neighborhoods. Reference metadata covers 3,233 of 4,751 papers, from bibliographic data of 9 September 2026, before the corpus was limited to AI papers. Numbers label the seven largest groups of sources cited together.

C DETAILED MECHANISM AND EVIDENCE PROFILES

Coding and populations. This appendix uses the coding of the mechanism cards completed by 28 August 2026, restricted to the systematic corpus (Section 3): 2,847 open-coded papers and 2,891 closed-field papers. Appendix A.6 lists the population behind each profile.

The complete facet catalogue supports the main mechanism analysis (Section 5); complementary evidence profiles support the workflow analysis (Section 4).

C.1 Category Maps and Mechanism Examples

The eighteen category maps cover all 110 mechanism facets in five families. Branches group design operations that systems may combine. Schematics explain the choices; source figures show cited implementations. Variations are recorded implementations, not settings available in every system. Map order and size carry no frequency or effectiveness scale.

Search the full definitions. Links beneath each map open its source mechanisms.

C.1.1 Explanation and visibility. What can a person inspect and understand?

FC-C01 / Explanation and visibility

Explaining system behaviour

Which aspect of the system is made inspectable?

Makes a system's outputs, behaviour, uncertainty or current state inspectable, so people can understand what it is doing and judge its results.



DESIGN CHOICES	RETAINED DISTINCTIONS	
A particular result	FC001 output rationale exposed	FC002 feature attribution for an output
Process and sequence	FC006 live process-state visibility	FC008 stepwise reasoning or execution trace
Model and internals	FC003 global model-behaviour summary	FC007 internal or latent state surfaced
Reliability and limits	FC004 uncertainty and confidence exposed	FC005 capability and knowledge-boundary disclosure

Recorded form · FC001: Rationale for a specific machine output (VA001).

Recorded form · FC002: Feature or factor contribution to an outcome (VA002).

Figure 15: Explaining system behaviour. Which aspect of the system is made inspectable? Separate the explanation of one result, the running process, the model across cases, and the limits of what it can establish. Category definitions; Browse category mechanisms. Recorded forms: Rationale for a specific machine output; Feature or factor contribution to an outcome.

FC-C01 / SOURCE ILLUSTRATION**Explaining system behaviour****PAPER ILLUSTRATION · FC006**

1. Intended route overlaid on the road 2. Detected obstacle marked in the scene

Live overlays expose the vehicle's planned route and detected objects in the road scene.

ExplAIn Yourself! Transparency for Positive UX in Autonomous Driving · Figure 2, PDF page 4

Schneider et al.; CC BY; cropped; numbered annotations added by this review.

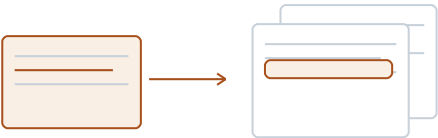
Figure 16: Explaining system behaviour: source illustration. Figure 2, PDF page 4 in [110]. Live overlays expose the vehicle's planned route and detected objects in the road scene. Inspect the illustrated mechanism.

FC-C02 / Explanation and visibility

Sources and evidence

How can a result be traced or checked?

Connects a result to its sources, supporting evidence or underlying cases, so people can trace where it came from and check its basis.



DESIGN CHOICES

RETAINED DISTINCTIONS

Origin and derivation

FC009

output-to-source provenance link

FC012

analytic-history provenance capture

Underlying cases

FC010

aggregate-to-instance drill-down

Evidence for a claim

FC011

evidence presented for verification

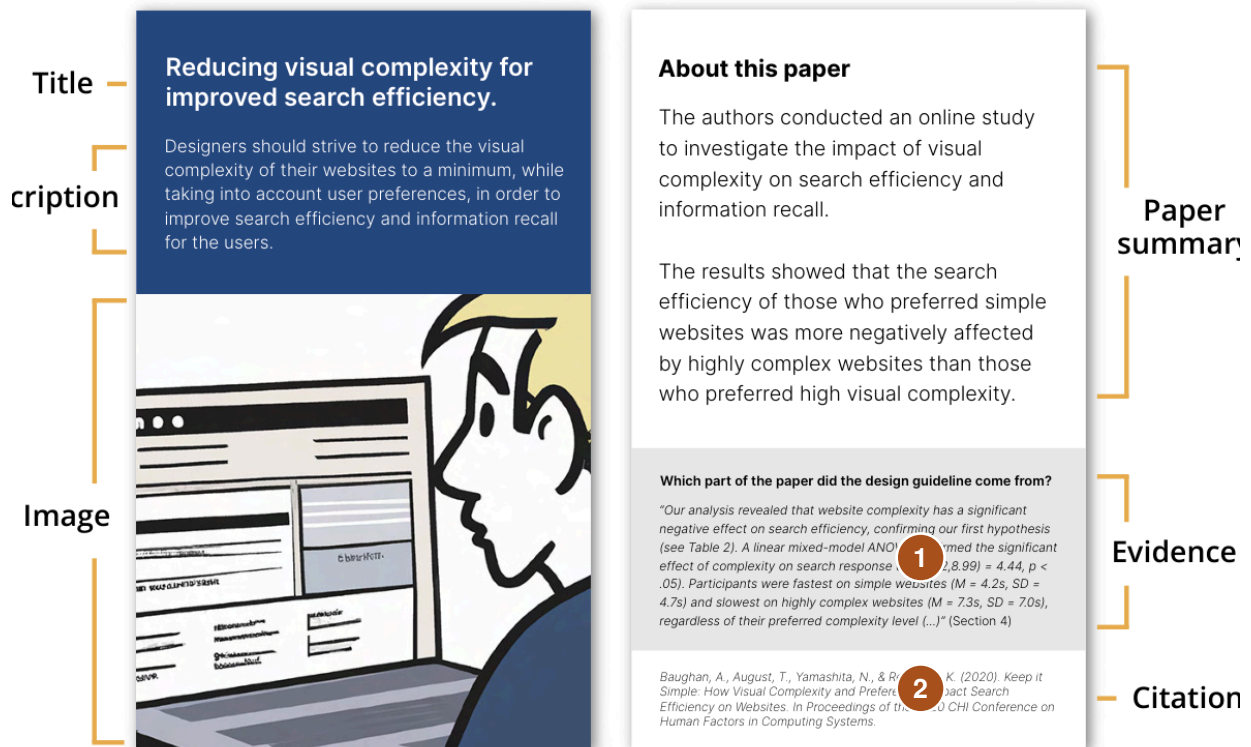
Recorded form · FC010: Drill-down from summary to underlying instances (VA012).

Figure 17: Sources and evidence. How can a result be traced or checked? Tracing origin and derivation, opening underlying cases, and assessing evidence answer different checking questions. Category definitions; Browse category mechanisms. Recorded forms: Drill-down from summary to underlying instances.

FC-C02 / SOURCE ILLUSTRATION

Sources and evidence

PAPER ILLUSTRATION · FC009



1. Supporting passage 2. Citation back to the paper

The design card places a supporting source passage and its citation beside the implication.

From Paper to Card: Transforming Design Implications with Generative AI · Figure 1, PDF page 2

Shin et al.; CC BY; cropped; numbered annotations added by this review.

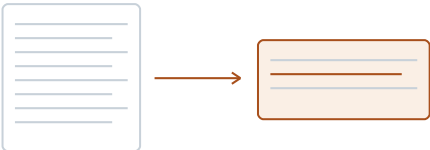
Figure 18: Sources and evidence: source illustration. Figure 1, PDF page 2 in [115]. The design card places a supporting source passage and its citation beside the implication. Inspect the illustrated mechanism.

FC-C03 / Explanation and visibility

Summarising and labelling content

How is content condensed, interpreted or recast?

Condenses or interprets content through summaries, labels, profiles or a change of representation, so people can grasp what it means.



DESIGN CHOICES

RETAINED DISTINCTIONS

Condense content



FC107

summarisation and condensation

Characterise an item



FC105

machine-generated semantic labelling

FC109

entity or concept profile

Change its form



FC108

semantic transformation into another form

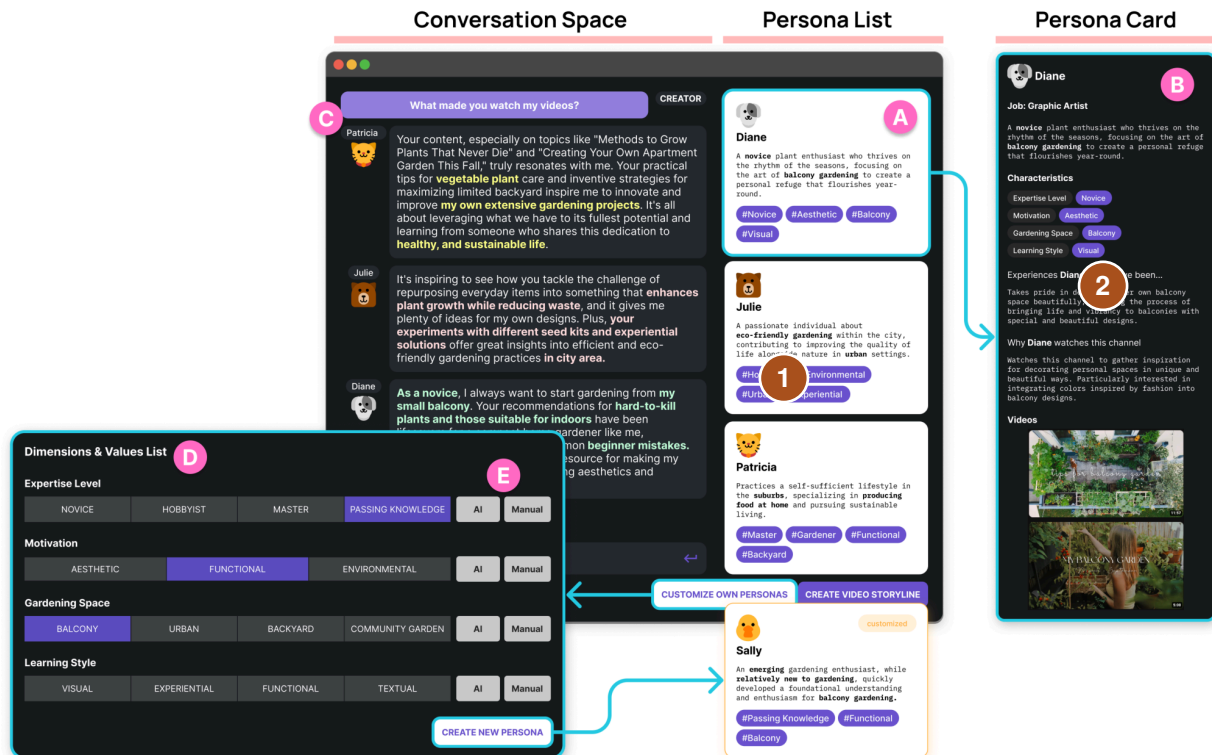
Recorded form · FC107: Condensed summary of a single item (VA097).

Figure 19: Summarising and labelling content. How is content condensed, interpreted or recast? Distinguish reducing a body of content, characterising an item, and changing representational kind. Category definitions; Browse category mechanisms. Recorded forms: Condensed summary of a single item.

FC-C03 / SOURCE ILLUSTRATION

Summarising and labelling content

PAPER ILLUSTRATION · FC109



1. Compact audience profiles 2. Expanded persona details

Audience segments become profile cards, with detailed experiences and motivations available on selection.

Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas · Figure 2, PDF page 6

Choi et al.; CC BY; cropped; numbered annotations added by this review.

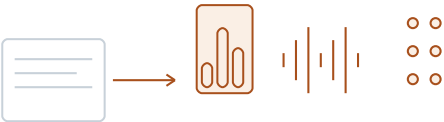
Figure 20: Summarising and labelling content: source illustration. Figure 2, PDF page 6 in [29]. Audience segments become profile cards, with detailed experiences and motivations available on selection. Inspect the illustrated mechanism.

FC-C04 / Explanation and visibility

Visual and sensory representation

Through which form can a person perceive meaning?

Expresses meaning through visual form, language, sound, touch or physical action, including translations and adaptations that make it accessible.



DESIGN CHOICES

RETAINED DISTINCTIONS

Perceptual carrier

FC085	FC083	FC086
visual encoding and graphical mapping	non-visual sensory encoding	embodied or physical enactment

Translation of meaning

FC084	FC089
cross-modal translation	verbal description of perceptual content

Adaptation and reframing

FC088	FC087
accessible or simplified presentation	metaphorical or symbolic re-representation

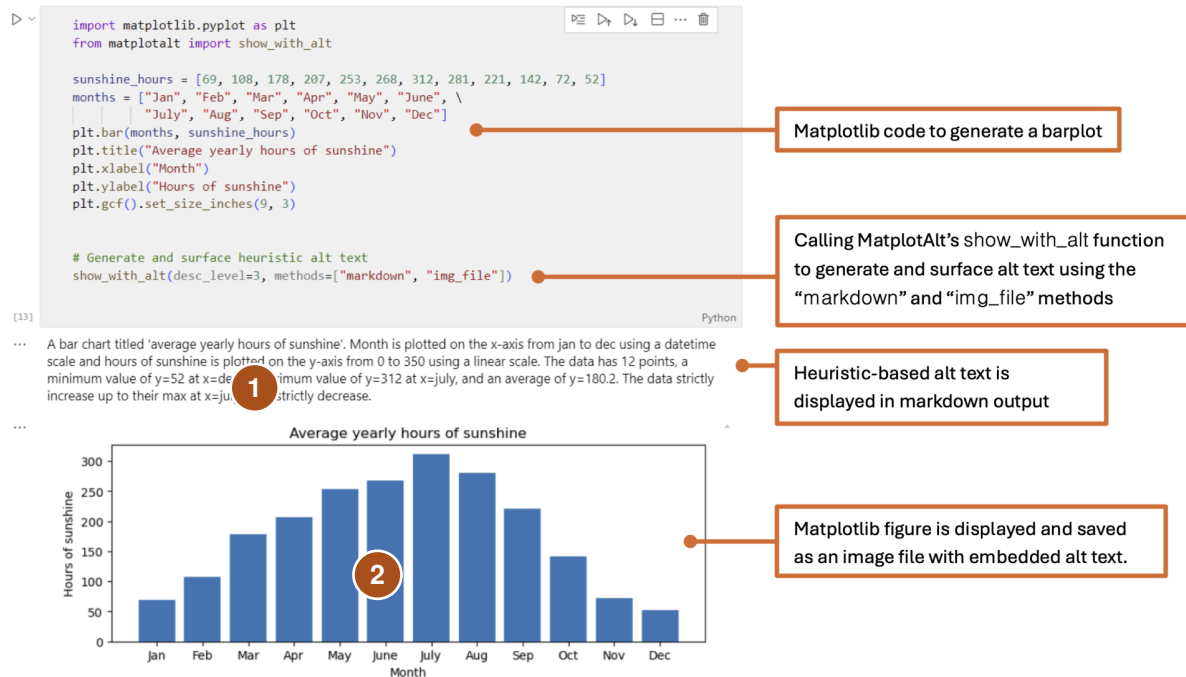
Recorded form · FC083: Spoken or audio rendering of content (VA071).

Figure 21: Visual and sensory representation. Through which form can a person perceive meaning? Separate the perceptual carrier, translation between carriers, and adaptations that change how the meaning can be grasped. Category definitions; Browse category mechanisms. Recorded forms: Spoken or audio rendering of content.

FC-C04 / SOURCE ILLUSTRATION

Visual and sensory representation

PAPER ILLUSTRATION · FC088



1. Description surfaced as text 2. The same chart remains visible

A notebook presents a chart and its generated description together; the library also supports alternative output placements.

Matplotlib: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks · Figure 1, PDF page 1

Nylund et al.; CC BY; cropped; numbered annotations added by this review.

Figure 22: Visual and sensory representation: source illustration. Figure 1, PDF page 1 in [93]. A notebook presents a chart and its generated description together; the library also supports alternative output placements. Inspect the illustrated mechanism.

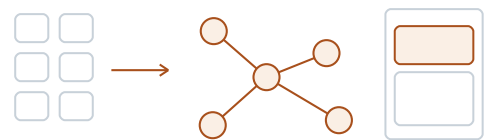
C.1.2 Organisation and comparison. How can a person organise and compare material?

FC-C05 / Organisation and comparison

Organising and structuring

Which relationships give material its structure?

Arranges information into relationships, groups, levels or sequences, so people can understand its structure and move between the whole and its parts.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Links and containment	FC015 relational or network map		FC013 hierarchical or nested organisation
Groups and categories	FC016 similarity grouping and clustering	FC017 categorisation scheme or controlled vocabulary	FC106 human semantic coding and categorisation
Whole, parts and detail	FC014 overview or aggregate gist	FC018 decomposition into parts	FC019 multiscale detail and semantic zoom
Position and alignment	FC020 temporal arrangement of material	FC021 spatial arrangement of material	FC022 cross-representation correspondence
Quantitative structure	FC023 distributional or statistical structure display		

Recorded form · FC015: Relational graph or dependency network (VA019).

Figure 23: Organising and structuring. Which relationships give material its structure? Relations, category membership, granularity, position and statistical structure organise different aspects of the same material and may be combined. Category definitions; Browse category mechanisms. Recorded forms: Relational graph or dependency network.

FC-C05 / SOURCE ILLUSTRATION

Organising and structuring

PAPER ILLUSTRATION · FC018



6. Research question node 7. Linked solution node

An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes.

Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback · Figure 4, PDF page 8

Pu et al.; CC BY; cropped; numbered annotations added by this review.

Figure 24: Organising and structuring: source illustration. Figure 4, PDF page 8 in [101]. An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes. Inspect the illustrated mechanism.

FC-C06 / Organisation and comparison

Comparing alternatives

What is placed in a common comparison frame?

Places alternatives, cases or states in a common frame so people can examine their similarities, differences and trade-offs.



DESIGN CHOICES

RETAINED DISTINCTIONS

Separate candidates

FC024

side-by-side alternative comparison

Changed states or versions

FC025

contrastive or counterfactual comparison

FC028

change-over-time comparison

Standards and populations

FC026

benchmark or peer-referenced comparison

FC027

subgroup or population comparison

Recorded form · FC024: Side-by-side comparison of alternatives (VA023).

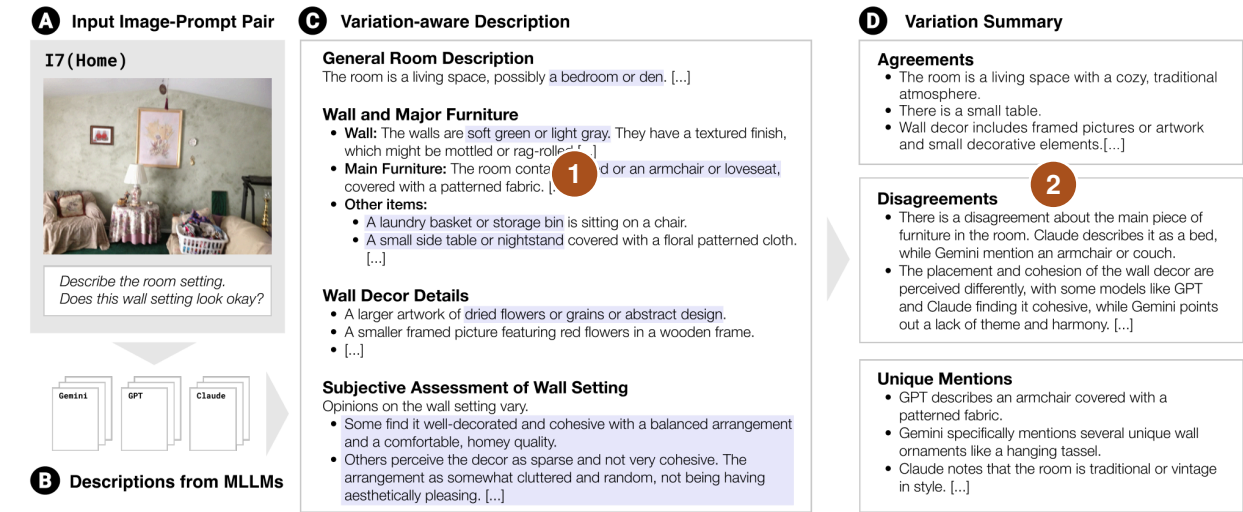
Recorded form · FC025: Before-and-after or version comparison (VA025).

Figure 25: Comparing alternatives. What is placed in a common comparison frame? The comparand distinguishes candidate selection, variation of the same object, and comparison against groups or reference points. Category definitions; Browse category mechanisms. Recorded forms: Side-by-side comparison of alternatives; Before-and-after or version comparison.

FC-C06 / SOURCE ILLUSTRATION

Comparing alternatives

PAPER ILLUSTRATION · FC024



1. Detailed merged description 2. Agreements and disagreements

The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary.

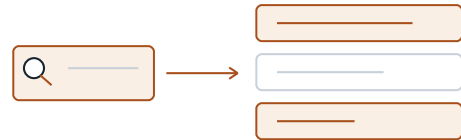
Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions · Figure 2, PDF page 5

Chen et al.; CC BY; cropped; numbered annotations added by this review.

Figure 26: Comparing alternatives: source illustration. Figure 2, PDF page 5 in [26]. The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary. Inspect the illustrated mechanism.

FC-C07 / Organisation and comparison**Search, filtering and ranking****How is relevant material found and brought into view?**

Helps people find relevant material and control what is in view through search, navigation, filters, rankings and scope choices.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Define the request****FC030**

query construction and search specification

FC033

analytic scope setting

Select and order**FC029**

relevance filtering of a corpus

FC032

ranked or prioritised surfacing

Navigate and look up**FC034**

navigation and wayfinding through structure

FC031

on-demand lookup of related information

Recorded form · FC030: Filter, query or search over a set (VA028).

Figure 27: Search, filtering and ranking. How is relevant material found and brought into view? Defining the request, selecting or ordering results, and moving through material are distinct retrieval operations. Category definitions; Browse category mechanisms. Recorded forms: Filter, query or search over a set.

FC-C07 / SOURCE ILLUSTRATION

Search, filtering and ranking

PAPER ILLUSTRATION · FC030

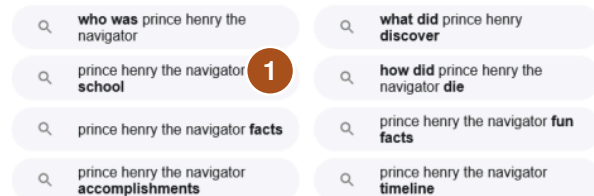
Searches related to: "logo design"

[online logo design](#) [design your own logo](#) [logo design tips](#) [logo design ideas](#)

Searches related to amazon diversity

[amazon diversity officer](#) [amazon black employee network](#)
[amazon diversity 2016](#) [amazon workforce login](#)
[amazon diversity report 2016](#) [amazon supplier diversity program](#)
[amazon supplier diversity registration](#) [amazon workforce phone number](#)

Related searches



1. Follow-on query formulations

Related-search suggestions provide alternative formulations for continuing a search.

From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features · Figure 16, PDF page 5

Oliveira et al.; CC BY; cropped; numbered annotations added by this review.

Figure 28: Search, filtering and ranking: source illustration. Figure 16, PDF page 5 in [94]. Related-search suggestions provide alternative formulations for continuing a search. Inspect the illustrated mechanism.

FC-C08 / Organisation and comparison**Notes, history and memory****What is kept available for continuing the work?**

Keeps information and work history available through notes, annotations, memory aids and replay, so people can revisit and continue their work.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Express and capture****FC077**

externalising internal structure into an artifact

FC080

capture of activity as it happens

Mark and arrange**FC078**

annotation onto material

FC082

reorganisable working workspace

Recall and revisit**FC079**

external memory and recall cue

FC081

replay and history navigation

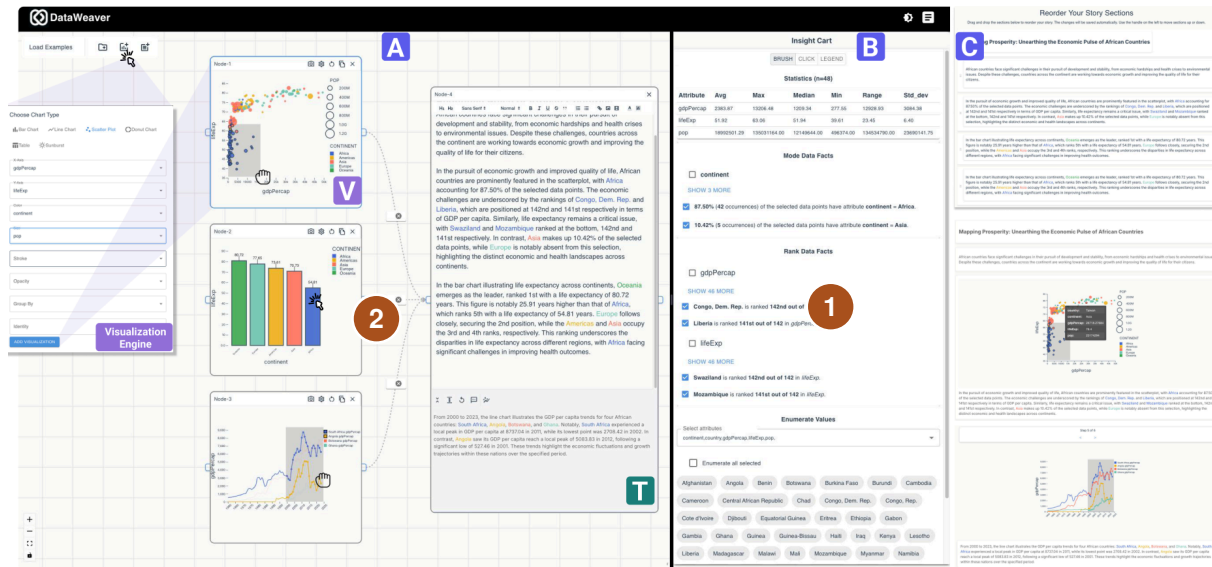
Recorded form · FC078: Annotation and comment capture on content (VA048).

Figure 29: Notes, history and memory. What is kept available for continuing the work? Separate making or capturing a record, arranging material at hand, and returning to an earlier state. Category definitions; Browse category mechanisms. Recorded forms: Annotation and comment capture on content.

FC-C08 / SOURCE ILLUSTRATION

Notes, history and memory

PAPER ILLUSTRATION · FC082



1. Facts held in the Insight Cart 2. Connected story components

An Insight Cart holds candidate facts for later selection into a connected data story.

DATAWEAVER: Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text ·

Figure 2, PDF page 5

Fu et al.; CC BY; cropped; numbered annotations added by this review.

Figure 30: Notes, history and memory: source illustration. Figure 2, PDF page 5 in [37]. An Insight Cart holds candidate facts for later selection into a connected data story. Inspect the illustrated mechanism.

C.1.3 User control and expression. What can a person express, change or try?

FC-C09 / User control and expression

Goals and constraints

How does a person specify what the system should do?

Lets people state what they want through goals, rules, criteria, examples or structured inputs that the system can act on.



DESIGN CHOICES

RETAINED DISTINCTIONS

Words and demonstrations

FC053

natural-language or conversational specification

FC054

example or demonstration specification

Structured and executable input

FC057

structured input form or schema

FC058

authoring a formal or executable representation

Conditions and criteria

FC055

explicit constraint or rule specification

FC056

criteria and rubric specification

Recorded form · FC055: Rule, criterion or rubric authoring (VA042).

Figure 31: Goals and constraints. How does a person specify what the system should do? Distinguish conveying intent, providing a structured or executable specification, and declaring constraints or evaluation criteria. Category definitions; Browse category mechanisms. Recorded forms: Rule, criterion or rubric authoring.

FC-C09 / SOURCE ILLUSTRATION**Goals and constraints****PAPER ILLUSTRATION · FC054**☐ **Show only Comments labeled as "Remove"**

30 / 30 labeled examples; 14 example(s) labeled as "Remove".

Those people who pull out guns just because they get upset are called gang banger thug losers. 99% of their guns are illegally possessed. Legal gun owners aren't taking out their guns during an argument at parties. Remove Keep

So I should get rid of my gun because some idiot killed. Remove Keep

Idiot sandwich 🥪 Remove Keep

Load More Examples Create your Filter

1. Keep or remove this example 2. Train the filter from labels

Situated keep/remove judgments supply examples of the content a person wants the classifier to retain.

End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting · Figure 2, PDF page 6

Wang et al.; CC BY; cropped; numbered annotations added by this review.

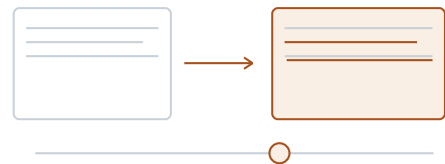
Figure 32: Goals and constraints: source illustration. Figure 2, PDF page 6 in [127]. Situated keep/remove judgments supply examples of the content a person wants the classifier to retain. Inspect the illustrated mechanism.

FC-C10 / User control and expression

Editing and steering

Where can a person intervene in the work?

Lets people change a representation, revise an output or redirect a process through edits, settings, parameters and reversible actions.



DESIGN CHOICES

RETAINED DISTINCTIONS

Object and output

FC047

direct manipulation of a semantic object

FC048

editable machine output

Settings and quantities

FC050

configuration of system behaviour

FC049

continuous parameter or dial control

Direction and commitment

FC051

steering an ongoing process

FC052

reversibility and safe experimentation

Recorded form · FC049: Parameter, threshold or weight controls (VA039).

Figure 33: Editing and steering. Where can a person intervene in the work? Editing an object, configuring behaviour, steering the current process and controlling commitment operate at different points. Category definitions; Browse category mechanisms. Recorded forms: Parameter, threshold or weight controls.

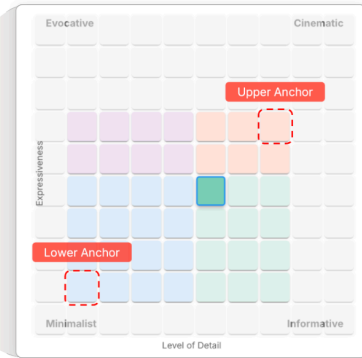
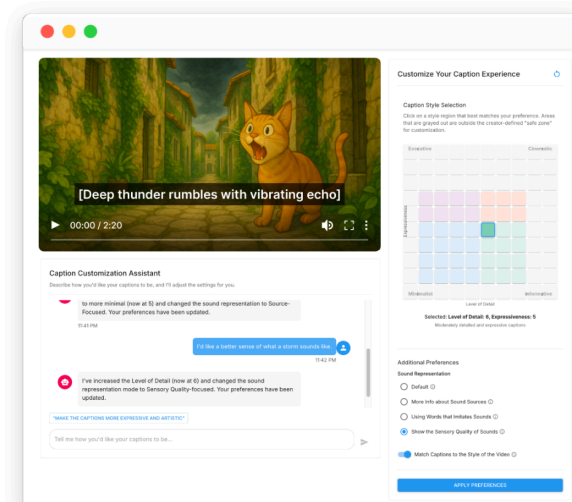
FC-C10 / SOURCE ILLUSTRATION

Editing and steering

PAPER ILLUSTRATION · FC050

1

Overall interface



2

Style Grid for *Level of Detail* (x-axis) and *Expressiveness* (y-axis). Creator-defined anchors—shown as red dots—specify the bounds of the caption transformation space.

Caption Preference Assistant

Describe how you'd like your captions to be, and I'll adjust the settings for you.

to more minimal (now at 5) and changed the sound representation to Source-Focused. Your preferences have been updated.

11:41 PM

I'd like a better sense of what a storm sounds like.

11:42 PM

I've increased the Level of Detail (now at 6) and changed the sound representation mode to Sensory Quality-focused. Your preferences have been updated.

MAKE THE CAPTIONS MORE EXPRESSIVE AND ARTISTIC

Tell me how you'd like your captions to be...

Sound Representation

- ☐ Default ⓘ
- ☐ More Info about Sound Sources ⓘ
- ☐ Using Words that Imitates Sounds ⓘ
- ☒ Show the Sensory Quality of Sounds ⓘ
- ☐ Match Captions to the Style of the Video ⓘ

3

Controls for *Sound Representation* and *Genre/Style Alignment*.

4

Interface where users set preferences by chatting with an AI assistant using natural language prompts.

5. Choose how the sound is represented

Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material.

CapTune: Adapting Non-Speech Captions With Anchored Generative Models · Figure 4, PDF page 8

Huang et al.; CC BY; cropped; numbered annotations added by this review.

Figure 34: Editing and steering: source illustration. Figure 4, PDF page 8 in [52]. Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material. Inspect the illustrated mechanism.

FC-C11 / User control and expression

Generating and choosing alternatives

How are possibilities offered and shaped?

Provides candidate outputs, continuations or a space of possibilities that people can explore, constrain and select from.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Present a choice space	FC035 candidate or option set to choose from	FC038 option-space landscape or design space
Generate or continue	FC036 generated alternatives to explore	FC110 completion and continuation of content
Bound generation	FC037 user-constrained generation	

Recorded form · FC036: Gallery of generated alternatives to choose among (VA033).
Recorded form · FC110: Completion or continuation of what the person began (VA036).

Figure 35: Generating and choosing alternatives. How are possibilities offered and shaped? Separate presenting a choice space, producing new content, and bounding the generation process. A continuation is not necessarily a set of alternatives. Category definitions; Browse category mechanisms. Recorded forms: Gallery of generated alternatives to choose among; Completion or continuation of what the person began.

FC-C11 / SOURCE ILLUSTRATION

Generating and choosing alternatives

PAPER ILLUSTRATION · FC110

Double-click the snippet to edit it directly. **THIS IS SUGGESTED TEXT**

Adjust role

Explain interaction × ⓘ

Adjust formatting

heading drophead body **note**

Get recommendations

Highlight interactive elements and guide the audience's interaction flow using moderately complex sentences. Use imperative sentences to instruct users on actions they can perform. Depending on the complexity, aim for 10–25 words, a readability level appropriate for someone with a 6th to 10th grade education, and a word variety of 50–70%.

Word count **5**

Word variety **40%** ⓘ

Readability score **69** ⓘ

Your text is likely to be understood by someone who has at least a 8th-grade education.

Generate text

[↻ Re-generate](#) [✂ Shorten](#) [📖 Simplify](#)

Add context to incorporate into generation

What should we consider when generating this text?

[🗑 Delete snippet](#)

1. Regenerate, shorten or simplify

A text snippet exposes explicit actions for generating a revised version while the author continues composing.

Plume: Scaffolding Text Composition in Dashboards · Figure 5, PDF page 11

Lisnic et al.; CC BY; cropped; numbered annotations added by this review.

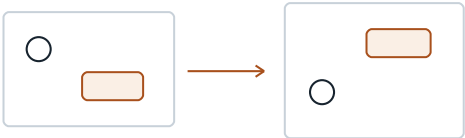
Figure 36: Generating and choosing alternatives: source illustration. Figure 5, PDF page 11 in [71]. A text snippet exposes explicit actions for generating a revised version while the author continues composing. Inspect the illustrated mechanism.

FC-C12 / User control and expression

What-if exploration and rehearsal

What can be tried before acting in the real setting?

Lets people try hypothetical changes, probe a system's behaviour or practise in a simulated setting and inspect what happens.



DESIGN CHOICES

RETAINED DISTINCTIONS

Probe the system



FC098

system-behaviour probing

Construct a scenario



FC097

scenario construction and what-if exploration

Rehearse an activity



FC096

simulated environment for rehearsal

Recorded form · FC096: Practice, rehearsal or simulation environment (VA058).

Figure 37: What-if exploration and rehearsal. What can be tried before acting in the real setting? Separate learning how a system responds, constructing a hypothetical world, and rehearsing a person's activity. Category definitions; Browse category mechanisms. Recorded forms: Practice, rehearsal or simulation environment.

FC-C12 / SOURCE ILLUSTRATION

What-if exploration and rehearsal

PAPER ILLUSTRATION · FC096



1. Dialogue history 2. Teacher question entry

Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history.

Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents · Figure 1, PDF page 3

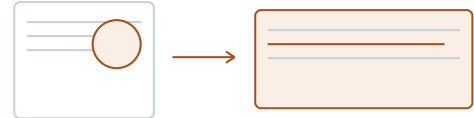
Barrett et al.; CC BY; cropped; numbered annotations added by this review.

Figure 38: What-if exploration and rehearsal: source illustration. Figure 1, PDF page 3 in [9]. Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history. Inspect the illustrated mechanism.

C.1.4 Guidance and feedback. How is the next action or reflection supported?

FC-C13 / Guidance and feedback**Recommendations and guidance****What helps a person decide where to look or what to do?**

Directs attention or suggests an action, resource or procedure, including corrective advice, consequence previews and warnings.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Notice and hold back****FC043**

attention direction and salience cueing

FC044

risk alert or warning

Preview a consequence**FC046**

prospective consequence preview

Choose an action or resource**FC040**

next-step recommendation

FC042

item or content recommendation

Perform or repair**FC045**

procedural how-to instruction

FC041

corrective or remediation guidance

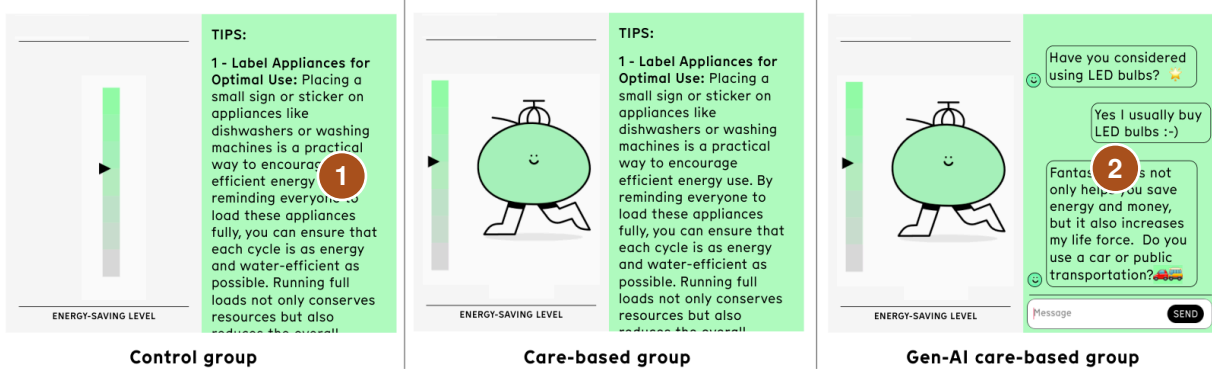
Recorded form · FC046: Preview of the consequence before committing (VA037).

Figure 39: Recommendations and guidance. What helps a person decide where to look or what to do? Attention and warnings, previews, recommendations, and procedural or corrective guidance contribute different information to an action. Category definitions; Browse category mechanisms. Recorded forms: Preview of the consequence before committing.

FC-C13 / SOURCE ILLUSTRATION

Recommendations and guidance

PAPER ILLUSTRATION · FC040



1. Fixed tips 2. Conversational condition

The study contrasts fixed energy-saving tips with a conversational condition that elicits household context.

Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment · Figure 6, PDF page 6

Berney et al.; CC BY; cropped; numbered annotations added by this review.

Figure 40: Recommendations and guidance: source illustration. Figure 6, PDF page 6 in [12]. The study contrasts fixed energy-saving tips with a conversational condition that elicits household context. Inspect the illustrated mechanism.

FC-C14 / Guidance and feedback**Feedback and checking****What basis is provided for judging the work?**

Shows how work is progressing or how a result meets a goal, criterion or reference. It includes feedback on attempts, problem diagnosis, quality assessments and checks.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Attempt and progress****FC065**

formative outcome feedback

FC066

progress and status tracking

Problem, quality and status**FC063**

Identifying problems in the work

FC064

evaluative judgment or quality rating

FC067

correctness or status marking

Check against a reference**FC068**

verification and checking against a reference

Recorded form · FC063: Error and failure diagnosis (VA007).

Figure 41: Feedback and checking. What basis is provided for judging the work? Separate feedback about an attempt or progress, appraisals and status marks, and an active check against a reference. Category definitions; Browse category mechanisms. Recorded forms: Error and failure diagnosis.

Feedback and checking

(a) Search bar

(b) Relevant paragraphs with descriptive section headings

(c) Explore similar paragraphs to this

(d) Highlighted dissimilar unexplored references

(e) Low-lighted previously-encountered information

(f) selected reference with paper details open below the paragraph text

Palani et al.; CC BY; cropped; numbered annotations added by this review.

Figure 42: Feedback and checking: source illustration. Figure 1, PDF page 2 in [96]. Persistent coverage bars and local unread-reference counts make progress through the literature visible. Inspect the illustrated mechanism.

FC-C15 / Guidance and feedback**Instruction and paced help****What help is available, and when is it available?**

Provides explanations, examples and orientation, and controls when or how much help is available through staged disclosure, learning paths or changing support.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Explain, exemplify and orient****FC101**

conceptual explanation of domain meaning

FC102

worked example or exemplar

FC103

orientation and onboarding

Change amount or release**FC099**

graduated or faded support

FC100

progressive disclosure of detail

FC039

deliberate restriction or withdrawal of support

Sequence learning sessions**FC104**

curriculum and learning-path structuring

Recorded form · FC099: Graduated help that escalates or fades by response (VA093).

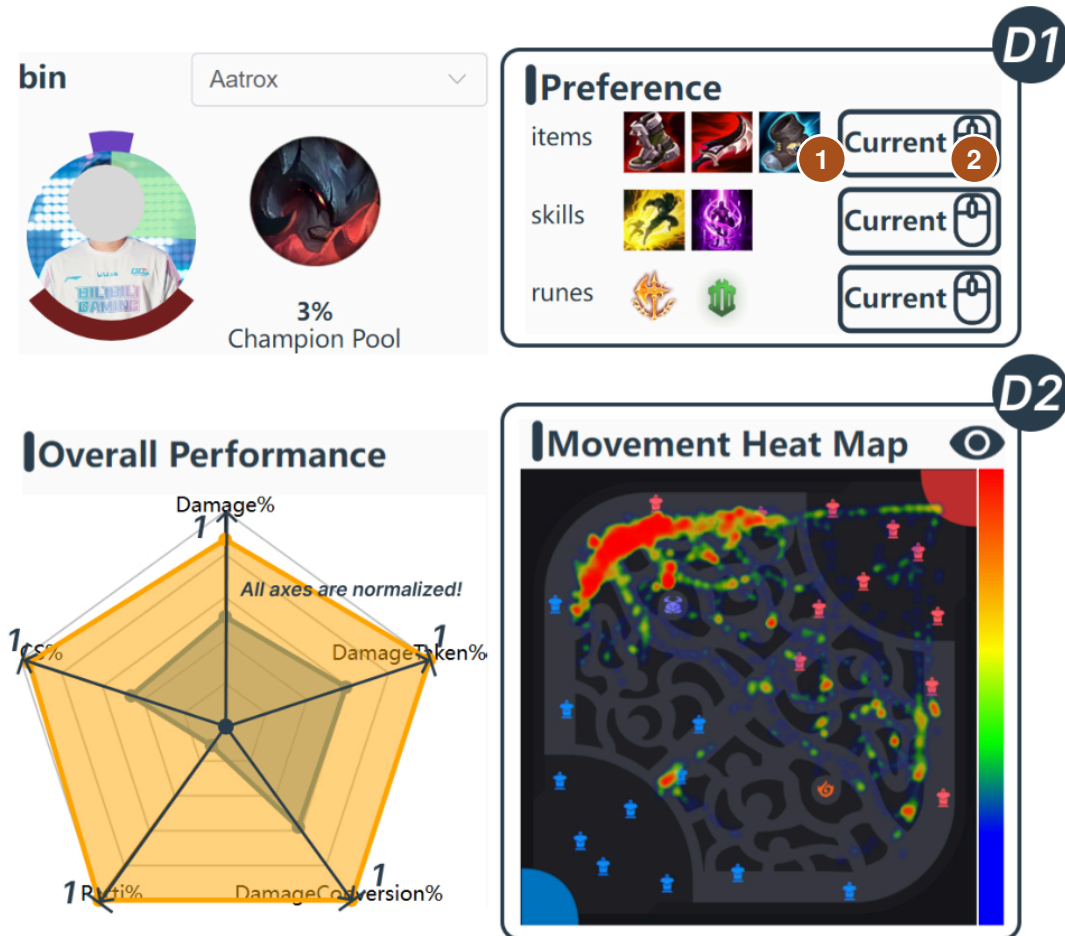
Recorded form · FC102: Worked or contrastive examples as instruction (VA057).

Figure 43: Instruction and paced help. What help is available, and when is it available? Instructional content, the amount or timing of support, and progression across sessions are distinct design choices. Fading is not automatically evidence of learning. Category definitions; Browse category mechanisms. Recorded forms: Graduated help that escalates or fades by response; Worked or contrastive examples as instruction.

FC-C15 / SOURCE ILLUSTRATION

Instruction and paced help

PAPER ILLUSTRATION · FC102



1. Historical equipment choices 2. Switch to current choices

Historical equipment, skill and rune choices can be inspected beside controls revealing the current match.

StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports · Figure 8, PDF page 12

Ma et al.; CC BY; cropped; numbered annotations added by this review.

Figure 44: Instruction and paced help: source illustration. Figure 8, PDF page 12 in [76]. Historical equipment, skill and rune choices can be inspected beside controls revealing the current match. Inspect the illustrated mechanism.

FC-C16 / Guidance and feedback

Structuring the workflow

How are stages, responsibilities and continuity arranged?

Organises work into stages, loops, roles and decision points, with ways to coordinate actions, hand work over, recover from problems and resume later.



DESIGN CHOICES

RETAINED DISTINCTIONS

Sequence and recurrence

—	FC069 staged or phased workflow	FC070 iterative revision loop
---	---	---

Responsibility and routing

—	FC071 human decision checkpoint or gate	FC072 division of labour across agents	FC074 orchestration and routing of actions
---	---	--	--

Continuity and recovery

—	FC073 handoff of work state	FC075 interaction repair and recovery	FC076 resumable and persistent work state
---	---------------------------------------	---	---

Recorded form · FC075: System move that clarifies, confirms or repairs (VA038).

Figure 45: Structuring the workflow. How are stages, responsibilities and continuity arranged? Separate progression through work, allocation of actions and authority, and continuity across interruptions or handoffs. Category definitions; Browse category mechanisms. Recorded forms: System move that clarifies, confirms or repairs.

FC-C16 / SOURCE ILLUSTRATION

Structuring the workflow

PAPER ILLUSTRATION · FC074

Controller Agent Prompts

[Stage Decision Prompt]

"Stage 0"
Once you've collected the student's information...
...

"Stage 1"
In the "Problem Discovery" phase, the student cannot proceed if:
(1) Fewer than 3-5 relevant problems are submitted.
(2) Problem statements lack scenario, subject, or

"Stage 2"
In the "Information Collection" phase, the student proceed if:
(1) Completed relevant information search....
(2) Written a status report on the...

"Stage 3"
In the "Problem Definition" phase, the student proceed if:
1
(1) Submitted one problem statement ...
(2) The statement includes scenario ...

"Stage 4"
In the "Solution Ideation" phase, the student proceed if:
(1) Proposed three feasible ...
(2) Each solution includes ...

"Stage 5"
In the "Solution Evaluation" phase, the student proceed if:
(1) Evaluated the three solutions ...
(2) The report must include at least ...

"Stage 6"
In the "Solution Implementation" phase, the student proceed if:
(1) Complete implementation plan ...
(2) The report includes background on the ...

Currently in phase: Stage.Name The current phase is defined as Stage.Description, examples are Stage.Example
The next phase is (Stage+1).Name, defined as (Stage+1).Description, examples are (Stage+1).Example
Based on the dialogue information, determine whether to proceed to the next phase. If staying in the current phase, set the output parameter Stage to (stage); otherwise, set the output parameter Stage to (stage+1).

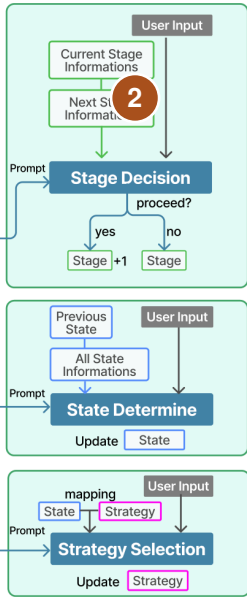
[State Determine Prompt]

If not moving to the next phase, the student's status may be: State.Name State.Description
The student's previous status list is State. Remove all '0' and '1'; and based on the new dialogue record, update and output the student's status list state.
If moving to the next phase, the student's status will be "start," and update and output the student's status list as state=[0].

[Strategy Selection Prompt]

The current student statuses and corresponding strategies are as follows: State.Name State.StrategyMapping Strategy.name Strategy.Description
Based on the dialogue information, output the most urgent problem ID as urge_state and the corresponding best strategy ID as strategy

Controller Agent Pipeline



1. Stage-specific completion criteria 2. Decision to advance or remain

Stage criteria and a controller decision determine whether the creative problem-solving workflow advances.

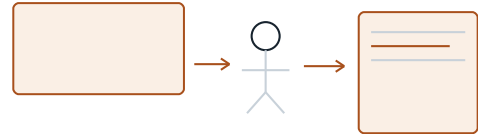
Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process · Figure 2, PDF page 9

Zha et al.; CC BY; cropped; numbered annotations added by this review.

Figure 46: Structuring the workflow: source illustration. Figure 2, PDF page 9 in [140]. Stage criteria and a controller decision determine whether the creative problem-solving workflow advances. Inspect the illustrated mechanism.

FC-C17 / Guidance and feedback**Questions and reflection prompts****Which part of a person's thinking is elicited?**

Asks people to express their goals, questions, explanations or reflections, making their own reasoning available to the work.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Needs and intentions****FC059**

intent and need elicitation

Account and reasoning**FC062**

narrative and reasoning elicitation

Questions and hypotheses**FC061**

inquiry and hypothesis prompting

Reflection on thinking**FC060**

reflective and metacognitive prompting

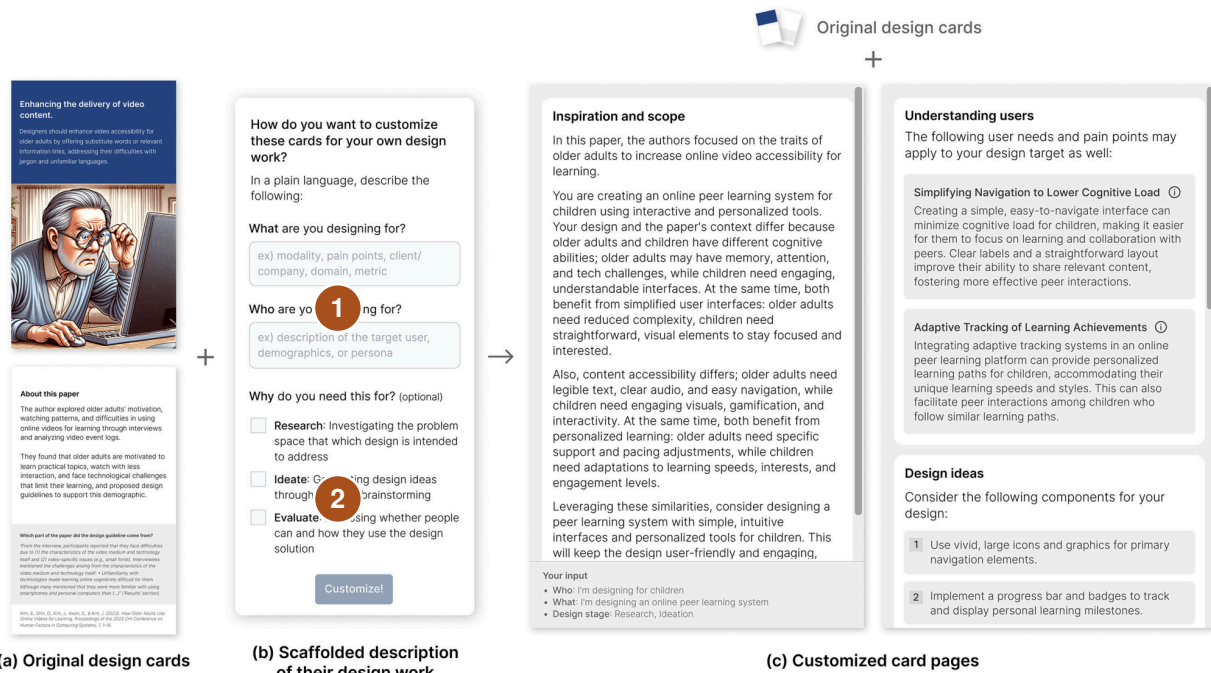
Recorded form · FC060: Reflection or self-assessment prompt (VA056).

Figure 47: Questions and reflection prompts. Which part of a person's thinking is elicited? The object of the prompt distinguishes needs, an account of reasoning, inquiry about the material, and reflection on one's own thinking. Category definitions; Browse category mechanisms. Recorded forms: Reflection or self-assessment prompt.

FC-C17 / SOURCE ILLUSTRATION

Questions and reflection prompts

PAPER ILLUSTRATION · FC059



1. Who and what is being designed 2. Select the design stage

A guided form elicits design context before the system customizes research cards.

What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts · Figure 1, PDF page 7

Shin et al.; CC BY; cropped; numbered annotations added by this review.

Figure 48: Questions and reflection prompts: source illustration. Figure 1, PDF page 7 in [114]. A guided form elicits design context before the system customizes research cards. Inspect the illustrated mechanism.

C.1.5 Collaboration and perspective. How can people share work and perspectives?

FC-C18 / Collaboration and perspective

Shared views and perspectives

How do contributions and viewpoints become shared?

Makes people's contributions and viewpoints available to one another through shared representations, structured discussion, role framing and control over what is disclosed.



DESIGN CHOICES

RETAINED DISTINCTIONS

Shared work and awareness



FC090

shared co-constructed representation

FC091

visibility of others' state and judgment

Viewpoints and exchange



FC092

perspective-taking and viewpoint framing

FC094

role or persona framing of the interaction

FC093

structured deliberation and perspective exchange

Control of disclosure



FC095

disclosure and privacy control

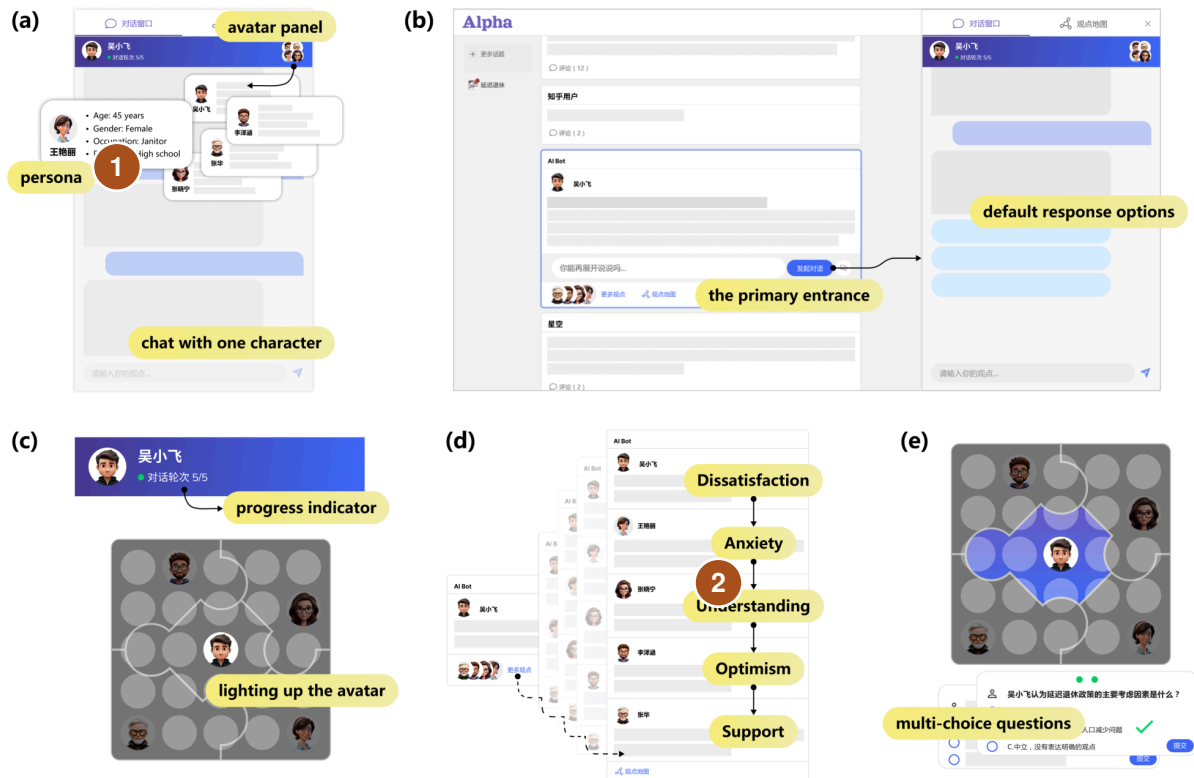
Recorded form · FC090: Shared workspace for joint work (VA065).

Figure 49: Shared views and perspectives. How do contributions and viewpoints become shared? Separate shared artifacts and awareness, the framing and exchange of perspectives, and control of disclosure. Category definitions; Browse category mechanisms. Recorded forms: Shared workspace for joint work.

FC-C18 / SOURCE ILLUSTRATION

Shared views and perspectives

PAPER ILLUSTRATION · FC092



1. Distinct stakeholder profiles 2. Progressive viewpoint sequence

Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure.

See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles · Figure 3, PDF page 7

Zhang et al.; CC BY; cropped; numbered annotations added by this review.

Figure 50: Shared views and perspectives: source illustration. Figure 3, PDF page 7 in [144]. Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure. Inspect the illustrated mechanism.

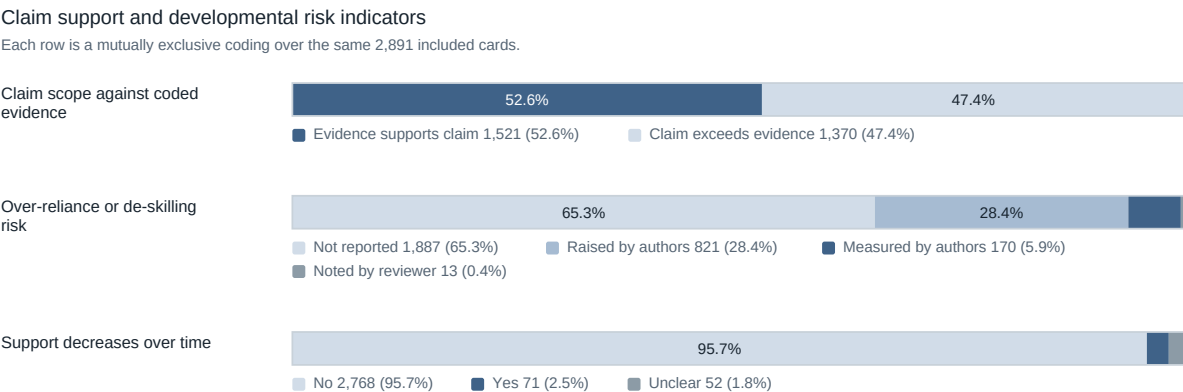


Figure 51: Claim scope exceeds coded evidence in nearly half of included cards, over-reliance is raised far more often than it is measured, and decreasing support is rare. Each row is a mutually exclusive coding over the same 2,891 included cards, and every segment prints its own count and share. A claim-evidence gap records the distance between a card’s stated claim and its linked evidence; it is not an assessment of the underlying paper. Examples, one or two cards per value: claim within evidence: InstructPipe [146].

Support profiles within learning-evidence groups

share of the row’s papers carrying each family; 2,761 papers coded in both fields

	what the interface makes reachable				
	Organisation and comparison	Guidance and feedback	Explanation and visibility	User control and expression	Collaboration and perspective
learning evidence					
Learning not measured n = 1,334 papers	73.8% 985	66.9% 893	64.2% 856	52.5% 701	11.4% 152
A proxy measured instead n = 1,039 papers	71.3% 741	74.1% 770	61.5% 639	50.0% 520	14.6% 152
Learning measured n = 258 papers	60.9% 157	82.9% 214	57.4% 148	41.5% 107	14.3% 37
Learning self-reported only n = 130 papers	71.5% 93	84.6% 110	63.8% 83	54.6% 71	20.0% 26
all papers in the matrix n = 2,761 papers	71.6%	72.0%	62.5%	50.7%	13.3%

colour: the row’s share against the column’s share over all papers; dark is 8 points or more above, pale 8 or more below

Figure 52: Guidance and feedback appear in 83% of capability-measured papers. Rows: the learning-evidence families, including the papers that measured no capability, with the number of papers in each. Columns: the facet families. Each cell is the share of the row’s papers carrying the column family, with the count; colour is the cell’s distance from the column’s share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

C.2 Complementary Evidence Profiles

The learning-by-family matrix uses the open-code classification and its 2,761-paper matched population. It is distinct from the closed-field learning count of 301 among 2,891 included papers.

C.3 Context and Evidence Profiles

The 2,891 closed-field papers record 2,071 empirical, 691 artifact, 123 methodological, four dataset, and two theoretical contributions. Contribution labels do not identify research methods or target users; study designs and participants are reported in Section 4.2.

In the design space, the domain mix shows no simple annual shift: technical, analytic, and creative professions account for 54.8% in 2019 and 55.5% in 2025, ranging from 53.4% to 61.6% across these years. Annual totals also reflect reading coverage and the incomplete 2026 publication window, so they do not estimate field-wide growth.

What is made reachable, against what the person still does

share of the row's papers in each column; 2,311 papers coded in both fields

what the interface makes reachable	what the person still does			
	The task the tool did not do	New work the tool created	The whole task kept by the person	Final authority kept by design
Organisation and comparison n = 1,639 papers	52.2% 856	29.1% 477	11.4% 187	7.1% 117
Guidance and feedback n = 1,679 papers	49.9% 838	30.1% 506	11.1% 187	8.8% 147
Explanation and visibility n = 1,452 papers	53.2% 772	28.0% 407	10.1% 146	8.7% 126
User control and expression n = 1,170 papers	42.0% 491	35.8% 419	14.2% 166	7.9% 93
Collaboration and perspective n = 311 papers	56.3% 175	19.0% 59	12.9% 40	11.9% 37
all papers in the matrix n = 2,311 papers	51.4%	30.2%	10.1%	8.2%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Figure 53: The profiles share an ordering, with different shares of new work. The matrix uses the 2,311 systematic papers with complete facet-value and remaining-work assignments. Rows: the facet families, with the number of papers carrying each. Columns: what the person still does. Each cell is the share of the row's papers whose remaining work falls in the column family, with the count. Papers can belong to several mechanism-family rows; each has one remaining-work summary category in this analysis. Colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

Study design is recorded for 2,071 papers: 691 within-subject designs (33.4%), 328 single-condition usability studies (15.8%), and 119 field deployments (5.7%). The 2,072 entries include one paper with two designs; percentages use papers. Usability studies reveal practical difficulties; deployments can expose work missed by short comparisons.

The participant field also covers 2,071 papers: mixed groups account for 540 (26.1%), target practitioners 499 (24.1%), and crowdworkers 251 (12.1%); 382 (18.4%) are unspecified. Participants need not represent a system's intended users, and missing descriptions do not establish broad applicability.

C.3.1 Most process shifts rest on an author's attribution. Of 8,045 process shifts across 2,891 papers, 6,102 (75.8%) are author-attributed, 1,746 (21.7%) use measured contrasts, and 197 are reviewer inferences. Baselines also differ: 1,940 use a control condition, 869 state none, and the remainder refer to prior practice or an author's assertion. Describing a changed workflow does not establish its effect size.

C.4 Mechanism Families and Remaining Work

The detailed profile complements the control/work comparison in Section 4.2.

D ANNUAL TRENDS AND VENUE RATES

These tests and counts support the comparisons in Section 4 and Section 5. Trends use AI-involved papers with usable codes for each field. Technology follows the AI a paper builds or uses, independently of publication year: generative AI includes hybrids and earlier neural generators; traditional AI/ML includes predictive, discriminative, and symbolic systems. The technology projection covers 3,447 classified papers; 639 remain unassigned. Each annual percentage uses papers with assigned technology and usable coding for that dimension in the same year. Count summaries combine 2019–2026; the annual lines separate the years. Alias disagreements are withheld for the affected field. Adjusted contrasts account for publication year and venue as described below.

D.1 Research-context trends

We model binary context membership with year and venue fixed effects, using venue-clustered standard errors with a finite-sample correction and Student's *t* reference distribution. The linear probability coefficient estimates average within-venue change in percentage points per year. Holm adjustment controls the family-wise error rate across the 16 directional tests. Intervals are pointwise 95% confidence intervals. Separate categorical-year tests check temporal variation and departures from a straight trend; these are available with the analysis data. Results describe the coded review frame.

Table 12: Control and candidate choice after omitting each major venue. Gaps are GenAI minus traditional AI/ML, adjusted for year and venue, in percentage points; intervals are pointwise 95% confidence intervals. The omission ranges are +11.9 to +14.3 for control and +7.5 to +10.0 for candidate choice. Holm correction includes all 51 outcomes in each fit.

Fit	Control and expression		Candidate choice	
	Gap [95% CI]	Holm p	Gap [95% CI]	Holm p
Primary	+13.7 [+8.6, +18.7]	< .001	+9.2 [+5.3, +13.1]	0.001
Omit CHI	+12.6 [+6.5, +18.7]	0.009	+7.5 [+3.8, +11.2]	0.011
Omit CSCW	+13.4 [+8.1, +18.7]	< .001	+9.1 [+4.9, +13.3]	0.004
Omit EMNLP	+14.2 [+9.1, +19.2]	< .001	+8.9 [+4.7, +13.1]	0.005
Omit IUI	+13.7 [+8.2, +19.3]	< .001	+9.6 [+5.5, +13.7]	0.002
Omit TVCG	+11.9 [+7.1, +16.8]	< .001	+8.4 [+4.2, +12.6]	0.011
Omit VIS	+14.3 [+9.4, +19.3]	< .001	+10.0 [+6.4, +13.6]	< .001

Table 11: Venue-adjusted annual changes in research contexts, in percentage points per year. Primary estimates use 2019–2025; the last column includes available 2026 papers. Sensitivity intervals and adjusted tests are provided online.

Context	Annual change [95% CI]	Adjusted p	With 2026
AI and machine learning practice	-0.59 [-1.18, +0.00]	0.725	-0.67
Data analysis and visualization	-0.34 [-0.89, +0.22]	1.000	-0.44
Software engineering and programming	-0.34 [-0.72, +0.04]	0.894	-0.27
Scientific research practice	+0.35 [-0.03, +0.73]	0.894	+0.40
Information seeking and knowledge work	+0.42 [+0.14, +0.69]	0.064	+0.36
Writing, reading and language	+0.22 [-0.10, +0.55]	1.000	+0.31
Design practice	-0.17 [-0.60, +0.26]	1.000	-0.06
Creative and media production	+0.16 [-0.34, +0.65]	1.000	+0.27
Education and learning	+0.11 [-0.33, +0.54]	1.000	+0.11
Health and clinical care	+0.29 [-0.24, +0.83]	1.000	+0.27
Accessibility and disability	+0.28 [+0.06, +0.50]	0.210	+0.04
Everyday life and personal decisions	-0.28 [-0.61, +0.04]	0.894	-0.10
Online platforms and civic information	-0.10 [-0.49, +0.28]	1.000	-0.07
Physical and embodied activity	-0.17 [-0.60, +0.25]	1.000	-0.09
Business, industry and work	+0.28 [-0.03, +0.60]	0.894	+0.14
Public sector, law and societal systems	-0.04 [-0.20, +0.12]	1.000	-0.12

D.2 Technology comparisons

We compare binary code membership using a linear probability model with technology, categorical publication year, and venue fixed effects. Primary estimates use 2019–2025; sensitivity models restrict the analysis to 2023–2025 or add partial 2026. Venue-clustered standard errors use a finite-sample correction and a Student t reference with one fewer degree of freedom than the number of venues. Intervals are pointwise 95% confidence intervals. Holm correction covers all 51 context, activity, family, and category contrasts jointly; each sensitivity window and venue omission is corrected separately. Six major venues (CHI, CSCW, EMNLP, IUI, TVCG, and VIS) are each omitted in turn. A major venue supplies at least 5% of the pooled primary-year coded denominator for workflow, families, or categories; this rule uses denominators rather than outcome values. Further descriptive checks use shared year–venue cells with at least one or at least five coded papers per technology. We vary weights by pooled paper count, within-cell overlap, equal cells, equal venues, and equal years. These weights target different populations; their spread is not a confidence interval. The online analysis lists all estimates, coding denominators, and common-support sizes.

Venue omissions. User control and expression occurs in 633/1,071 GenAI papers (59.1%) and 623/1,373 traditional AI/ML papers (45.4%) in 2019–2025. Choosing among candidates occurs in 209/1,069 GenAI papers (19.6%) and 160/1,369 traditional AI/ML papers (11.7%) in 2019–2025. Table 12 gives the primary and six venue-omission estimates. All six omission fits retain corrected support for both contrasts.

Secondary contrasts. Direct manipulation loses corrected support when omitting CHI (Holm-adjusted $p=0.117$). Comparison activity loses corrected support when omitting IUI (Holm-adjusted $p=0.066$). Feedback and assessment loses corrected support when omitting CHI (Holm-adjusted $p=0.195$). These patterns remain secondary.

Which papers and venues carry the contrast? Among papers with usable coding, GenAI-classified papers more often describe user control and expression (+13.7 percentage points after accounting for year and venue; 95% CI 8.6 to 18.7). On the same 2,022 papers in shared year–venue cells, the gap is +12.5 with overlap weights and +7.8 when venues receive equal weight. Candidate choice is more sensitive to

Table 13: Descriptive common-cell sensitivities. Each row gives GenAI minus traditional AI/ML prevalence in percentage points. The one-paper threshold retains 82.7% of the two coded populations; the five-paper threshold retains 40.9–41.0% of the two coded populations. Different weights and thresholds change the target population. These are point descriptions, not confidence intervals.

Weight	At least one per group		At least five per group	
	Control	Choice	Control	Choice
Pooled papers	+10.8	+5.4	+17.3	+10.1
Within-cell overlap	+12.5	+7.2	+17.1	+9.7
Equal cells	+8.2	+4.4	+16.6	+9.1
Equal venues	+7.8	-0.04	+15.1	+4.7
Equal years	+10.0	+6.2	+18.8	+13.6

Table 14: Mechanism-family trends, 2019–2025: 2,444 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	+1.16[−0.70, +3.03]	1.000	+1.41[−0.97, +3.79]	1.000
Organisation and comparison	+0.06[−1.73, +1.84]	1.000	+0.43[−2.08, +2.95]	1.000
User control and expression	−1.02[−2.44, +0.41]	0.938	+0.53[−1.17, +2.23]	1.000
Guidance and feedback	+1.58[+0.70, +2.47]	0.008	+1.87[−0.44, +4.19]	0.771
Collaboration and perspective	+1.10[+0.52, +1.68]	0.005	+2.24[+0.03, +4.46]	0.379

Table 15: Mechanism-family trends, 2023–2025: 1,390 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	−0.21[−5.64, +5.22]	1.000	+1.18[−6.50, +8.87]	1.000
Organisation and comparison	−2.88[−6.68, +0.93]	0.941	−2.82[−9.62, +3.99]	1.000
User control and expression	−3.32[−6.48, −0.16]	0.322	−0.60[−6.51, +5.31]	1.000
Guidance and feedback	+1.87[−2.73, +6.47]	1.000	+3.60[−3.74, +10.94]	1.000
Collaboration and perspective	+2.67[+1.20, +4.14]	0.007	+6.64[+1.90, +11.38]	0.066

weighting: on its 2,016 shared-cell papers, the corresponding gaps are +7.2 and -0.04 points. The GenAI excess disappears when each venue receives equal weight. Equal venue weighting gives sparse cells much of the weight.

Unusable coding. Allowing each missing membership to range independently from absent to present in all 2,961 technology-classified primary-year papers gives adjusted identification bounds of -7.8 to +29.8 points for control and -11.1 to +27.1 for choice. These bounds include both signs; they are not confidence intervals or likely missing-data scenarios. They do not cover coding errors, papers with unresolved technology, or papers outside the coded set.

Sparse cells and code richness. For candidate choice, cells with at most two papers of a technology account for only 6.8% of GenAI and 5.9% of traditional AI/ML papers, but receive 40.8% and 46.9% of their equal-venue weight. Standardizing both groups to the same pooled number of non-control categories leaves 19.7% versus 9.6% with at least two control categories (+10.1 points, all 2,438 jointly coded papers). Thus, simply recording more other categories does not account for the pattern.

D.3 Temporal change in mechanism families

We distinguish average annual change in family prevalence from annual change in the GenAI–traditional AI/ML difference. Both come from a linear probability model with technology, publication year, their interaction and venue fixed effects. Average change holds the technology mixture fixed at 43.75% GenAI (the original primary cohort’s reference share), separating it from shifts in the composition of the corpus. Positive gap change means movement toward a more positive GenAI difference; it need not enlarge the absolute gap. Venue-clustered intervals use finite-sample correction and a Student t reference. Holm correction covers all ten tests within each window; the 95% confidence intervals are pointwise. These are linear summaries of coded design prevalence, not tests of learning or causal technology effects. Partial 2026 is excluded.

D.4 Annual distributions

D.5 Learning overlays

These counts cover the 3,378 papers with usable category coding, of the 4,086 papers in the analysis at 26 September 2026, including papers without an assigned AI technology. Learning scope means the paper is in the learning review; measured means the paper measured learning explicitly. Learning scope and Measured are counted separately within each category; missing or conflicting labels are omitted from the corresponding column.

Table 16: Papers per mechanism category: all papers, papers in learning scope, and papers that measured learning. Counts are distinct papers; categories overlap.

Category	Papers	Learning scope	Measured
Explaining system reasoning	971	213	97
Linking to sources	675	119	33
Summarising and labelling content	414	107	37
Visual and multimodal form	464	155	54
Organising and structuring	1,372	293	93
Comparing alternatives	725	166	60
Search, filtering and ranking	769	177	70
Notes, history and memory	603	170	65
Specifying intent	815	161	55
Direct manipulation	736	138	55
Choosing among candidates	484	104	36
What-if and probing	169	79	32
Recommending next steps	986	330	133
Feedback and assessment	898	318	121
Instruction and paced help	681	347	154
Workflow structure	655	173	69
Prompting the person to reflect	496	242	90
Shared views and perspective-taking	405	163	57

D.6 Venue candidates and current inclusions

Table 17: Venue candidates and AI-involved inclusions in the 16 July 2026 frame, assessed through 26 September 2026. The 37,978 venue-year records represent 37,660 candidate identities. Inclusion (%) is Included divided by Initial candidates. Papers without AI, and papers whose identity or scope is unresolved, are not in Included. Venue memberships can overlap; group and overall totals count each identity once.

Venue	Initial candidates	Included	Inclusion%
HCI & interaction	17,436	2,546	14.6%
56.0% of included corpus			
CHI	7,639	1,147	15.0%
UIST	860	156	18.1%
IUI	687	317	46.1%
DIS	1,471	166	11.3%
CSCW	2,412	212	8.8%
C&C	301	44	14.6%
TOCHI	478	37	7.7%
TiS	230	95	41.3%
EICS	174	23	13.2%
HAI	214	26	12.1%
ECSCW	205	4	2.0%
GROUP	142	6	4.2%
ASSETS	387	59	15.2%
HRI	630	94	14.9%

Venue	Initial candidates	Included	Inclusion%
IVA	222	27	12.2%
CHIIR	312	40	12.8%
HCOMP	112	28	25.0%
IJHCS	960	65	6.8%
Design & creativity 2.2% of included corpus	2,502	102	4.1%
ICCC	422	28	6.6%
Design Science	232	15	6.5%
Design Studies	234	1	0.4%
IDETC	1,614	58	3.6%
Visualization 19.7% of included corpus	6,336	895	14.1%
VIS	1,772	349	19.7%
EuroVis	537	84	15.6%
TVCG	4,073	445	10.9%
VL/HCC	255	65	25.5%
Graphics 2.5% of included corpus	2,385	115	4.8%
SIGGRAPH	1,673	87	5.2%
SIGGRAPH Asia	712	28	3.9%
Learning & education 5.0% of included corpus	2,834	226	8.0%
L@S	181	34	18.8%
LAK	655	85	13.0%
AIED	1,168	41	3.5%
IJAIED	347	34	9.8%
EDM	483	32	6.6%
Software engineering 7.5% of included corpus	3,944	343	8.7%
CHASE	249	18	7.2%
ICSE	1,038	125	12.0%
FSE	1,297	88	6.8%
ASE	1,360	112	8.2%
AI & language 7.0% of included corpus	2,223	317	14.3%
ACL	731	133	18.2%
EMNLP	794	146	18.4%
RecSys	698	38	5.4%
Overall (distinct identities)	37,660	4,544	12.1%

E VOCABULARY GROWTH DURING READING

These curves show how each vocabulary grew as cards were read, so a reader can judge how complete the codes are. The vocabulary built from the first 3,295 mechanism cards gained 12 codes across the next 1,754 papers, and each new code was assigned to an existing category and family. All open fields were checked on 821 of those papers; the other batches checked mechanism facets, values, and domains (Figure 58).

E.1 Reading the Curves

We count distinct values after each batch of ten cards, in reading order, and scale each curve to its own final vocabulary. Codes, categories, and families are drawn where those levels exist; categories and families were assigned after coding, so they can reach their final size early in a curve. Reference lists in the lineage cards are open-ended, while their relation options are fixed by the card format. A flat segment means that no new label appeared in those cards, not that unread papers would add none.

E.2 Stream 1 mechanism cards: Interface and Outcome Fields

These grids show the variation the overview hides. Every panel states its field and final count, and lines are drawn only for the levels present in that field.

Research contexts: volume and annual share

3,266 coded papers with assigned AI technology - categories overlap

■ Traditional AI/ML

■ GenAI

* 2026 partial: through 16 July

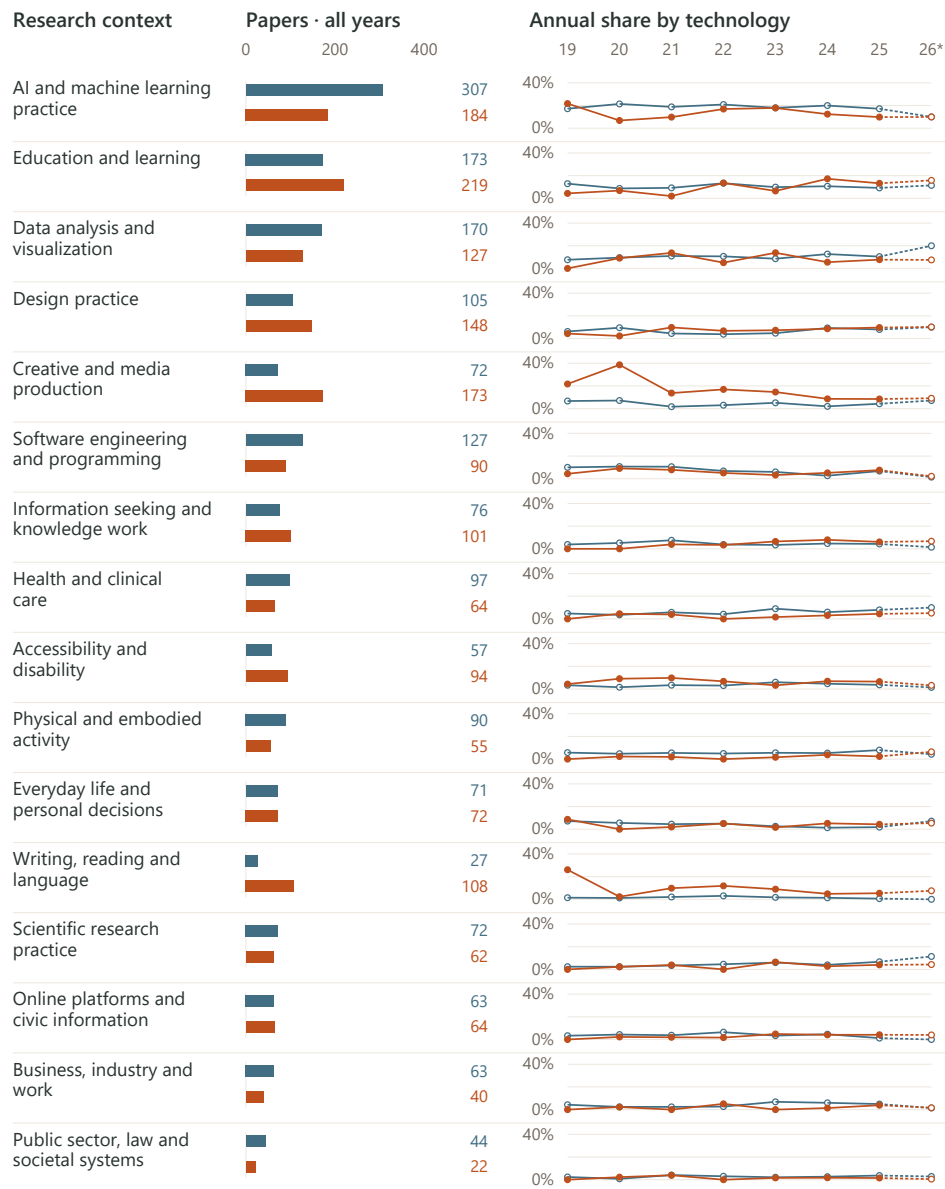


Figure 54: Research contexts by AI technology. Bars show paper counts across all years; lines show each context's share of its technology group per year, with the coded papers behind those shares below. Contexts overlap. Shares are not adjusted; Table 11 gives the venue-adjusted tests. Only papers with usable context coding and a known technology are included.

Workflow activities: volume and annual share

2,390 coded papers with assigned AI technology · categories overlap

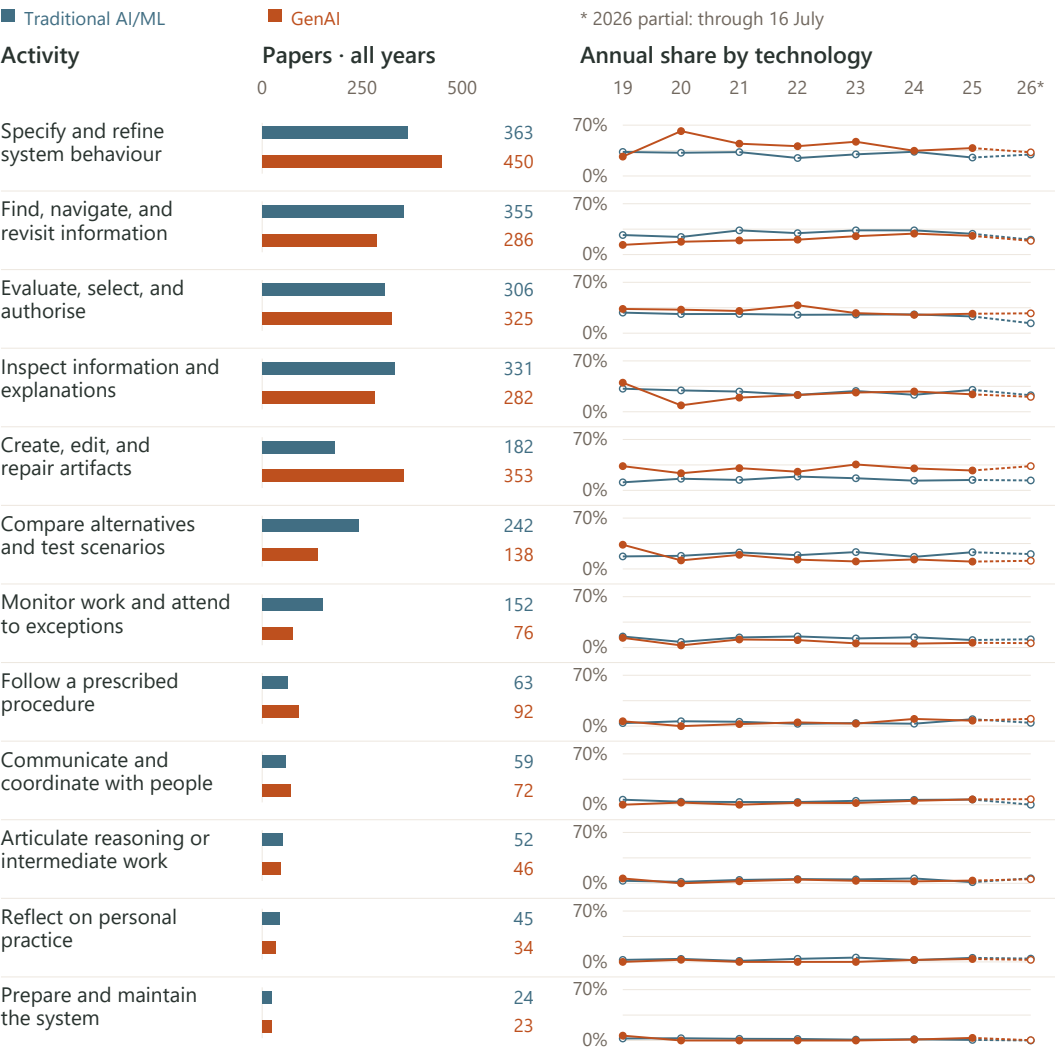


Figure 55: Workflow activities by AI technology. Bars show counts across all years; annual lines show shares within each technology group. Denominators count papers with usable activity codes and a known technology. Activities overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue.

Mechanism families: volume and annual share

2,845 coded papers with assigned AI technology · categories overlap

■ Traditional AI/ML

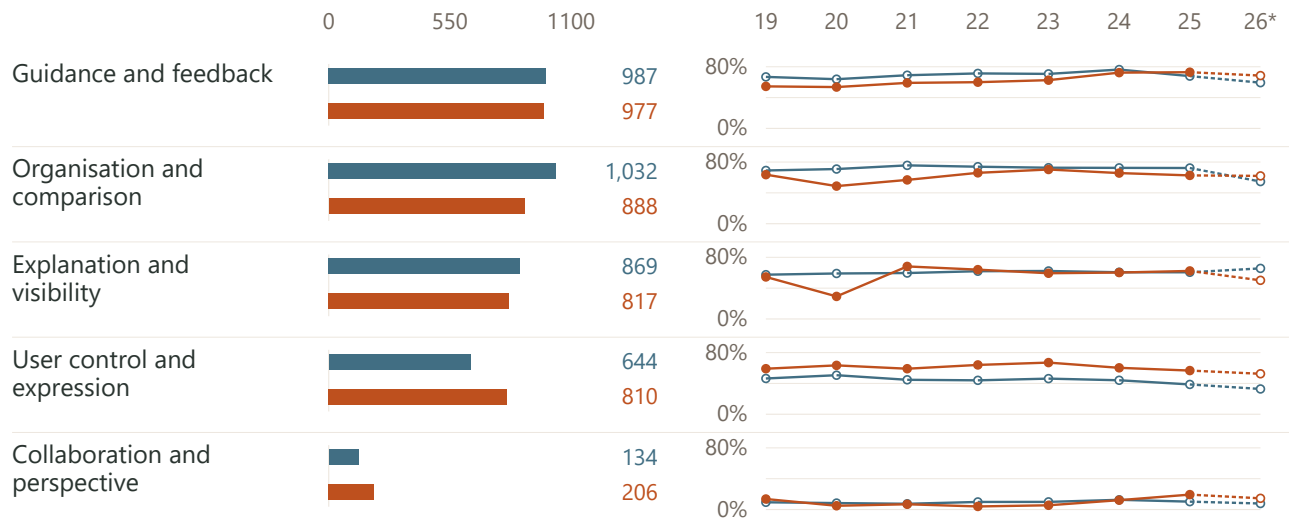
■ GenAI

* 2026 partial: through 16 July

Mechanism family

Papers · all years

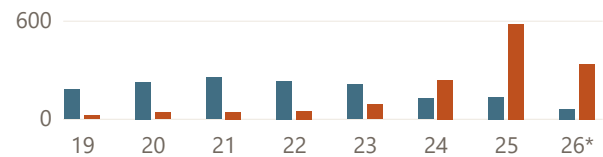
Annual share by technology



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

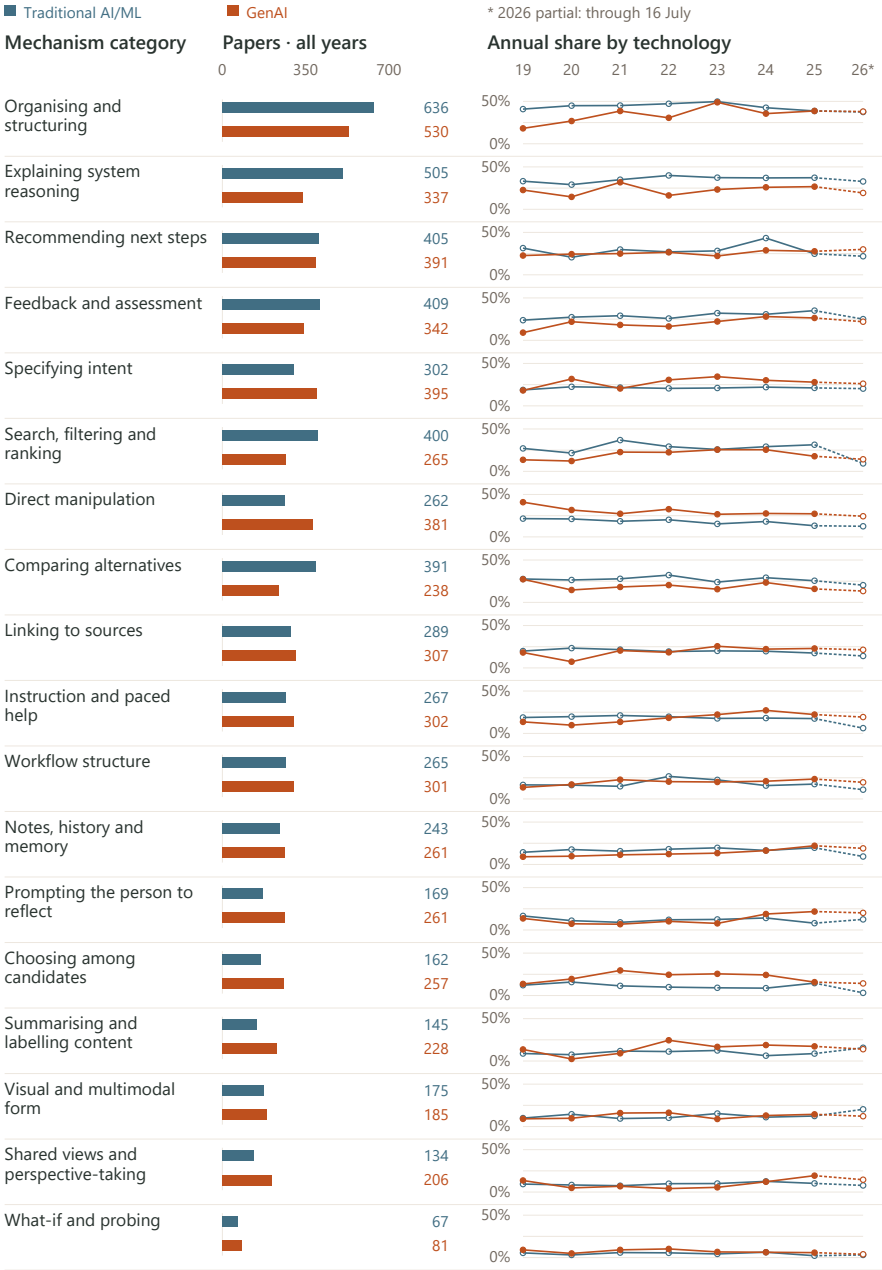


GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Figure 56: Mechanism families by AI technology. Bars encode counts across all years; lines compare annual shares within each technology group. The lower bars show annual coding denominators. Families overlap. Only papers with usable family codes and a known technology are included. Shares are not adjusted; Table 14 gives the venue-adjusted trends.

Mechanism categories: volume and annual share

2,839 coded papers with assigned AI technology · categories overlap



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

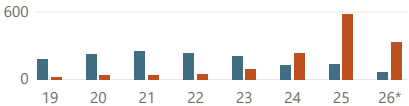
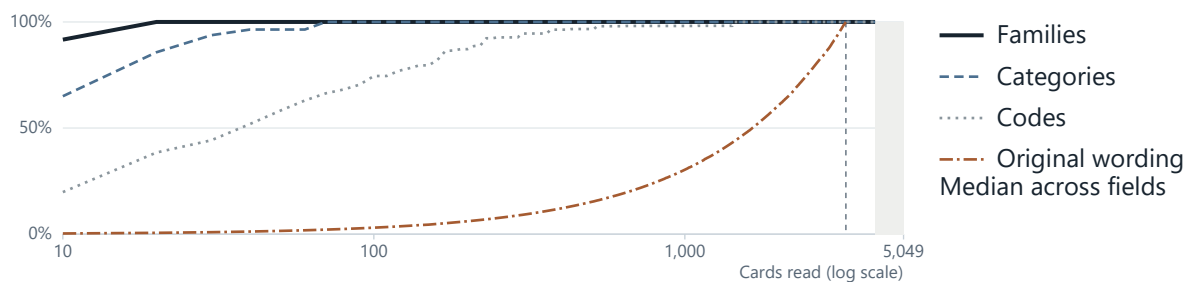


Figure 57: Mechanism categories by AI technology. Bars encode counts across all years; lines show annual shares within each technology group. Categories overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue. Only papers with usable category codes and a known technology are included.

Vocabulary growth across three card types

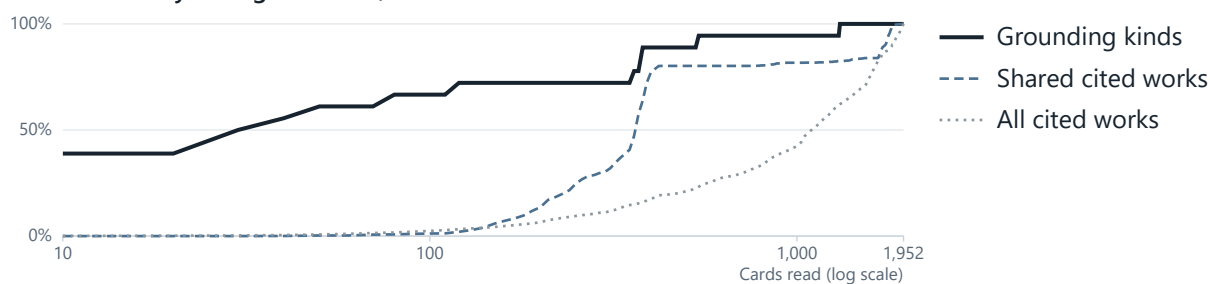
Share of the recorded vocabulary encountered; paper counts use a log scale

Stream 1 · Mechanism cards · 5,049



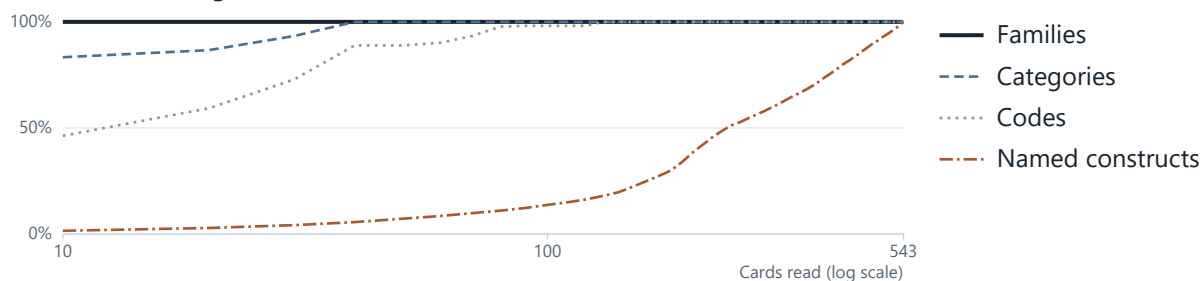
After 3,295: 12 new codes in 1,754 cards; field coverage varies in the shaded region.

Stream 1 · Theory-lineage cards · 1,952



Cited works remain an open vocabulary; grounding kinds are a fixed schema.

Stream 2 · Knowledge-claim cards · 543



Frozen coding covers 542 cards; named constructs cover all 543. Curves stop at their coverage.

Figure 58: Vocabulary growth across the three card types. Panels count cards in reading order on logarithmic axes: 5,049 mechanism cards, 1,952 theory-lineage cards, and 543 Stream 2 cards. The Stream 1 panels include cards of non-AI comparison papers. Shading marks uneven field coverage, and median curves stop before it.

Codes, categories and families reached, by cards read, every field



Figure 59: Stream 1, mechanism cards: vocabulary growth in each open field. Each panel separates codes, categories, and families where present, in reading order.

E.3 Stream 1 theory-lineage cards: Cited Sources and Their Roles

The curve covers the 1,952 newly written lineage cards; reused cards appear in the gallery but not in this sequence.

Stream 1 theory-lineage cards: references and relations

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Relations, positions and grounding kinds use fixed options: their flat tails are not evidence of open-code saturation.

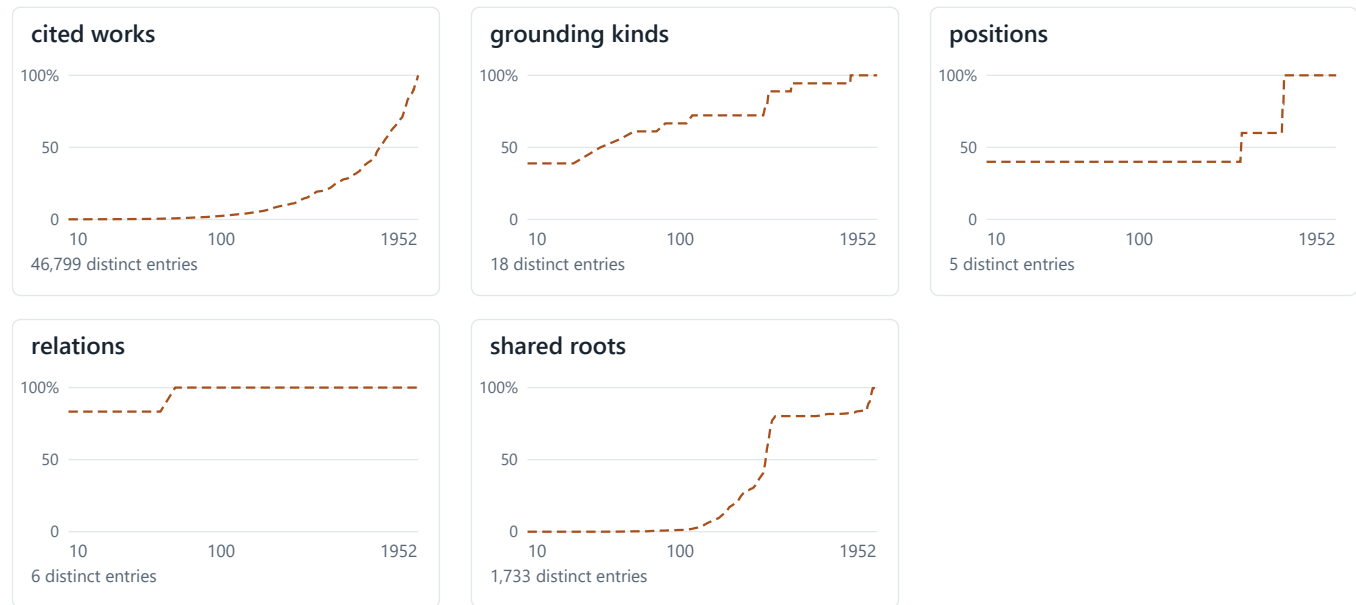


Figure 60: Stream 1, theory-lineage cards: reference lists keep growing after relation options are exhausted. Fixed options and open-ended lists are separate panels because they close in different ways.

E.4 Stream 2: Claims and Conditions

Stream 2: each coded field

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Dotted: raw values. Dashed: canonical codes. Solid: categories. Thick: families, where present.

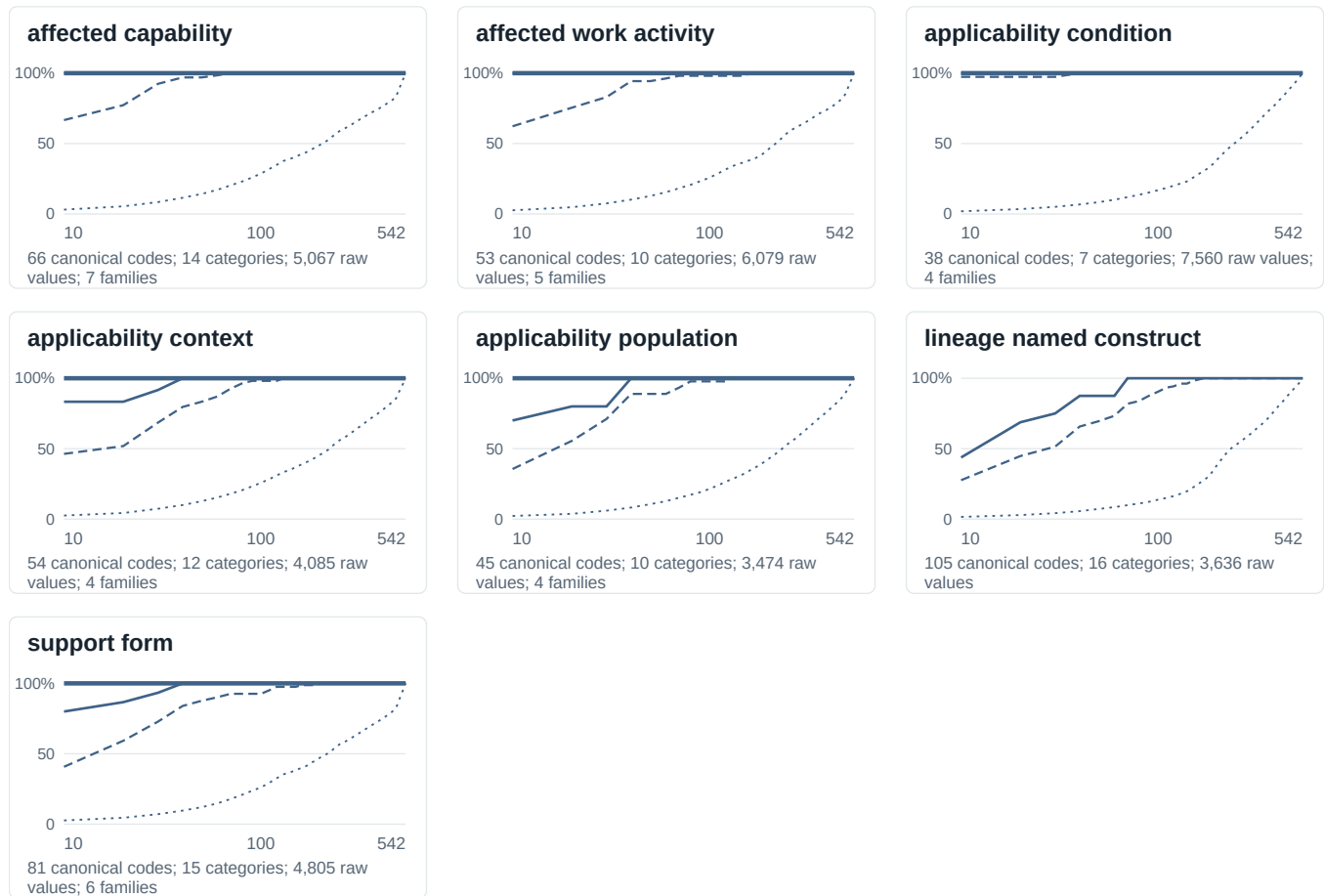


Figure 61: Stream 2: vocabulary growth differs by coded field. Codes, categories, and families are drawn only where that level exists.

Stream 2's seven open fields cover 542 cards and 35,699 observations. Of the 449 codes, 448 are used, and no new code entered in the last 16 coding batches. A flat curve can still hide a missing code: no population code names a studied unit that is or includes a machine, and 18 assignments flag this while using codes for people or artifacts.

E.5 New Codes Within the First Coding

Within the first coding of each open field, the last new code appeared between 2.5% and 48.7% of the way through the fields whose coding order was kept (Figure 62). Three fields still have low-confidence assignments for more than a quarter of their observations, so a vocabulary can stop growing and remain too coarse.

Where did the recorded vocabulary stop expanding?

Colour ends at the last new code. The pale remainder reuses codes already encountered.



■ Nine fields retain observation order; ticks show first-use counts within each tenth of that sequence.

■ Twelve fields retain batch order only; their bars locate the last batch that introduced a code.

This checks label reuse within the recorded sequence, not coverage of every meaningful distinction. Stable labels can still contain low-confidence or overly broad assignments.

Figure 62: New-code creation within each field of the first coding. Bars locate the last new code in the recorded coding sequence; ticks show new codes by tenth of the sequence. Where only batch order survives, the figure reports batch resolution.